

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12402523
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Stoessel; Philipp et al.

Metal complexes

Abstract

The present invention relates to iridium complexes suitable for use in organic electroluminescent devices, especially as emitters.

Inventors: Stoessel; Philipp (Frankfurt am Main, DE), Auch; ARmin (Darmstadt, DE)

Applicant: UDC IRELAND LIMITED (Dublin, IE)

Family ID: 1000008779756

Assignee: UDC IRELAND LIMITED (Dublin, IE)

Appl. No.: 18/320715

Filed: May 19, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230320191 A1	Oct. 05, 2023

Foreign Application Priority Data

EP	17206950	Dec. 13, 2017
----	----------	---------------

Related U.S. Application Data

continuation parent-doc US 16771537 US 11659763 WO PCT/EP2018/084094 20181210 child-doc US 18320715

Publication Classification

Int. Cl.: H10K85/30 (20230101); C07F15/00 (20060101); C09K11/06 (20060101); H10K50/11 (20230101); H10K101/10 (20230101)

U.S. Cl.:

CPC H10K85/342 (20230201); C07F15/0033 (20130101); C09K11/06 (20130101); C09K2211/1007 (20130101); C09K2211/1029 (20130101); C09K2211/1044 (20130101); C09K2211/185 (20130101); H10K50/11 (20230201); H10K2101/10 (20230201)

Field of Classification Search

CPC: H10K (85/342)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2015/0221877	12/2014	De Cola	546/4	C07F 15/0033
2015/0243910	12/2014	De Cola	N/A	N/A
2018/0026209	12/2017	Stoessel	252/519.2	C07F 15/008

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
107207550	12/2016	CN	N/A
2013243234	12/2012	JP	N/A
201422767	12/2013	TW	N/A
201700489	12/2016	TW	N/A
2014033044	12/2013	WO	N/A
2016124304	12/2015	WO	N/A
2017032439	12/2016	WO	N/A
2017092481	12/2016	WO	N/A

OTHER PUBLICATIONS

J. Li, et al., "Synthesis and characterization of cyclometalated Ir(III) complexes with pyrazolyl ancillary ligands", Polyhedron, 2004, pp. 419-428, vol. 23. cited by applicant

Primary Examiner: Young; William D

Attorney, Agent or Firm: Riverside Law LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This present application is a Continuation of U.S. patent application Ser. No. 16/771,537, filed Jun. 10, 2022, which is a U.S. national stage application under 35 U.S.C. § 371 of International Application No. PCT/EP2018/084094, filed on Dec. 10, 2018, which claims priority to European Patent Application No. 17206950.2, filed on Dec. 13, 2017, the entire disclosure of which are hereby incorporated by reference in its entirety for all purposes.

- (1) The present invention relates to iridium complexes suitable for use in organic electroluminescent devices, especially as emitters.
- (2) According to the prior art, triplet emitters used in phosphorescent organic electroluminescent devices (OLEDs) are iridium complexes in particular, especially bis- and tris-ortho-metallated complexes having aromatic ligands, where the ligands bind to the metal via a negatively charged carbon atom and an uncharged nitrogen atom or via a negatively charged carbon atom and an uncharged carbene carbon atom. Examples of such complexes are tris(phenylpyridyl)iridium(III) and derivatives thereof, and a multitude of related complexes, for example with 1- or 3-phenylisoquinoline ligands, with 2-phenylquinoline ligands or with phenylcarbene ligands. Even though complexes having polypodal ligands show advantages over complexes which otherwise have the same ligand structure except that the individual ligands therein do not have polypodal bridging, there is still a need for improvement here too. This lies particularly in the efficiency of luminescence of the compounds.
- (3) Also known are iridium complexes having three bidentate ligands, one of which is a pyrazolylborate ligand. These have a difficulty that the pyrazolylborate ligand has a tendency to hydrolytic breakdown in the course of synthesis (J. Li et al., Polyhedron 2004, 23, 419-428), and so it is not possible here to utilize the standard synthesis route via direct reaction of the chloro-bridged dimer with a pyrazolylborate ligand. Complexes of this kind show a tendency to hydrolysis in solution too. Further improvements are desirable here.
- (4) The problem addressed by the present invention is that of providing novel and especially improved metal complexes suitable as emitters for use in OLEDs.
- (5) It has been found that, surprisingly, this problem is solved by metal complexes with a hexadentate tripodal ligand having the structure described below and containing one or two pyrazolylborate ligands, which are of very good suitability for use in an organic electroluminescent device. The present invention therefore provides these metal complexes and organic electroluminescent devices comprising these complexes.

Description

- (1) The invention thus provides a compound of the formula (1)
- (2) ##STR00001##
- where the symbols used are as follows: L.sup.1 is a sub-ligand of the following formula (2) which coordinates to the iridium via the two nitrogen atoms identified by * and which is bonded to V via the dotted bond,
- (3) ##STR00002## where: A is the same or different at each instance and is CR or N, where not more than one A group per ring is N; R.sup.B is the same or different at each instance and is F, OR.sup.1, a straight-chain alkyl group having 1 to 10 carbon atoms or a branched or cyclic alkyl group having 3 to 10 carbon atoms, where the alkyl group in each case may be substituted by one or more R.sup.1 radicals, or an aromatic or heteroaromatic ring system which has 5 to 24 aromatic ring atoms and may be substituted in each case by one or more R.sup.1 radicals; at the same time, the two R.sup.B radicals together may also form a ring system; L.sup.2 is a bidentate, monoanionic sub-ligand which coordinates to the iridium via one carbon atom and one nitrogen atom or via two carbon atoms; L.sup.3 is a bidentate, monoanionic sub-ligand which coordinates to the iridium via one carbon atom and one nitrogen atom or via two carbon atoms, or is a sub-ligand of the above-detailed formula (2) which may be the same as or different from L.sup.1; V is a group of the formula (3), where the dotted bonds each represent the bonds to the sub-ligands L.sup.1, L.sup.2 and L.sup.3,
- (4) ##STR00003## X.sup.1 is the same or different at each instance and is CR or N; X.sup.2 is the same or different at each instance and is CR or N, or two adjacent X.sup.2 groups together are NR,

O or S, thus forming a five-membered ring; or two adjacent X^{sup.2} groups together are CR or N when one of the X^{sup.3} groups in the cycle is N, thus forming a five-membered ring; with the proviso that not more than two adjacent X^{sup.2} groups in each ring are N; X^{sup.3} is C at each instance in one cycle or one X^{sup.3} group is N and the other X^{sup.3} group in the same cycle is C, where the X^{sup.3} groups in the three cycles may be selected independently, with the proviso that two adjacent X^{sup.2} groups together are CR or N when one of the X^{sup.3} groups in the cycle is N; R is the same or different at each instance and is H, D, F, Cl, Br, I, N(R^{sup.1}).sub.2, OR^{sup.1}, SR^{sup.1}, CN, NO.sub.2, COOH, C(=O)N(R^{sup.1}).sub.2, Si(R^{sup.1}).sub.3, B(OR^{sup.1}).sub.2, C(=O)R^{sup.1}, P(=O)(R^{sup.1}).sub.2, S(=O)R^{sup.1}, S(=O).sub.2R^{sup.1}, OSO.sub.2R^{sup.1}, a straight-chain alkyl group having 1 to 20 carbon atoms or an alkenyl or alkynyl group having 2 to 20 carbon atoms or a branched or cyclic alkyl group having 3 to 20 carbon atoms, where the alkyl, alkenyl or alkynyl group may in each case be substituted by one or more R^{sup.1} radicals and where one or more nonadjacent CH₂ groups may be replaced by Si(R^{sup.1}).sub.2, C=O, NR^{sup.1}, O, S or CONR^{sup.1}, or an aromatic or heteroaromatic ring system which has 5 to 40 aromatic ring atoms and may be substituted in each case by one or more R^{sup.1} radicals; at the same time, two R radicals together may also form a ring system; R^{sup.1} is the same or different at each instance and is H, D, F, Cl, Br, I, N(R^{sup.2}).sub.2, OR^{sup.2}, SR^{sup.2}, CN, NO.sub.2, Si(R^{sup.2}).sub.3, B(OR^{sup.2}).sub.2, C(=O)R^{sup.2}, P(=O)(R^{sup.2}).sub.2, S(=O)R^{sup.2}, S(=O).sub.2R^{sup.2}, OSO.sub.2R^{sup.2}, a straight-chain alkyl group having 1 to 20 carbon atoms or an alkenyl or alkynyl group having 2 to 20 carbon atoms or a branched or cyclic alkyl group having 3 to 20 carbon atoms, where the alkyl, alkenyl or alkynyl group may in each case be substituted by one or more R^{sup.2} radicals and where one or more nonadjacent CH₂ groups may be replaced by Si(R^{sup.2}).sub.2, C=O, NR^{sup.2}, O, S or CONR^{sup.2}, or an aromatic or heteroaromatic ring system which has 5 to 40 aromatic ring atoms and may be substituted in each case by one or more R^{sup.2} radicals; at the same time, two or more R^{sup.1} radicals together may form a ring system; R^{sup.2} is the same or different at each instance and is H, D, F or an aliphatic, aromatic and/or heteroaromatic organic radical, especially a hydrocarbyl radical, having 1 to 20 carbon atoms, in which one or more hydrogen atoms may also be replaced by F; at the same time, the three bidentate ligands L^{sup.1}, L^{sup.2} and L^{sup.3}, apart from by the V group, may also be closed by a further bridge to form a cryptate.

(5) According to the invention, the ligand is thus a hexadentate tripodal ligand having three bidentate sub-ligands L^{sup.1}, L^{sup.2} and L^{sup.3}. “Bidentate” means that the particular sub-ligand in the complex coordinates or binds to the iridium via two coordination sites. “Tripodal” means that the ligand has three sub-ligands bonded to the V group or the group of the formula (3). Since the ligand has three bidentate sub-ligands, the overall result is a hexadentate ligand, i.e. a ligand which coordinates or binds to the iridium via six coordination sites. The expression “bidentate sub-ligand” in the context of this application means that L^{sup.1}, L^{sup.2} or L^{sup.3} would in each case be a bidentate ligand if the V group or the group of the formula (3) were not present. However, as a result of the formal abstraction of a hydrogen atom from this bidentate ligand and the attachment to the group of the formula (3), it is no longer a separate ligand but a portion of the hexadentate ligand which thus arises, and so the term “sub-ligand” is used therefor.

(6) The ligand of the compound of the invention thus has the following structure (LIG):

(7) ##STR00004##

(8) The bond of the ligand to the iridium may either be a coordinate bond or a covalent bond, or the covalent fraction of the bond may vary according to the sub-ligand. When it is said in the present application that the ligand or the sub-ligand coordinates or binds to the iridium, this refers in the context of the present application to any kind of bond from the ligand or sub-ligand to the iridium, irrespective of the covalent component of the bond.

(9) When two R or R^{sup.1} radicals together form a ring system, it may be mono- or polycyclic, and aliphatic, heteroaliphatic, aromatic or heteroaromatic. In this case, these radicals which

together form a ring system may be adjacent, meaning that these radicals are bonded to the same carbon atom or to carbon atoms directly bonded to one another, or they may be further removed from one another. For example, it is also possible for an R radical bonded to the X.sup.2 group to form a ring with an R radical bonded to the X.sup.1 group.

(10) The wording that two or more radicals together may form a ring, in the context of the present description, shall be understood to mean, inter alia, that the two radicals are joined to one another by a chemical bond with formal elimination of two hydrogen atoms. This is illustrated by the following scheme:

(11) ##STR00005##

(12) In addition, however, the abovementioned wording shall also be understood to mean that, if one of the two radicals is hydrogen, the second radical binds to the position to which the hydrogen atom was bonded, forming a ring. This shall be illustrated by the following scheme:

(13) ##STR00006##

(14) As described above, this kind of ring formation is possible in radicals bonded to carbon atoms directly adjacent to one another, or in radicals bonded to further-removed carbon atoms. Preference is given to this kind of ring formation in radicals bonded to carbon atoms directly bonded to one another.

(15) An aryl group in the context of this invention contains 6 to 40 carbon atoms; a heteroaryl group in the context of this invention contains 2 to 40 carbon atoms and at least one heteroatom, with the proviso that the sum total of carbon atoms and heteroatoms is at least 5. The heteroatoms are preferably selected from N, O and/or S. The heteroaryl group in this case preferably contains not more than three heteroatoms. An aryl group or heteroaryl group is understood here to mean either a simple aromatic cycle, i.e. benzene, or a simple heteroaromatic cycle, for example pyridine, pyrimidine, thiophene, etc., or a fused aryl or heteroaryl group, for example naphthalene, anthracene, phenanthrene, quinoline, isoquinoline, etc.

(16) An aromatic ring system in the context of this invention contains 6 to 40 carbon atoms in the ring system. A heteroaromatic ring system in the context of this invention contains 1 to 40 carbon atoms and at least one heteroatom in the ring system, with the proviso that the sum total of carbon atoms and heteroatoms is at least 5. The heteroatoms are preferably selected from N, O and/or S. An aromatic or heteroaromatic ring system in the context of this invention shall be understood to mean a system which does not necessarily contain only aryl or heteroaryl groups, but in which it is also possible for a plurality of aryl or heteroaryl groups to be interrupted by a nonaromatic unit (preferably less than 10% of the atoms other than H), for example a carbon, nitrogen or oxygen atom or a carbonyl group. For example, systems such as 9,9'-spirobifluorene, 9,9-diarylfuorene, triarylamine, diaryl ethers, stilbene, etc. shall thus also be regarded as aromatic ring systems in the context of this invention, and likewise systems in which two or more aryl groups are interrupted, for example, by a linear or cyclic alkyl group or by a silyl group. In addition, systems in which two or more aryl or heteroaryl groups are bonded directly to one another, for example biphenyl, terphenyl, quaterphenyl or bipyridine, shall likewise be regarded as an aromatic or heteroaromatic ring system.

(17) A cyclic alkyl, alkoxy or thioalkoxy group in the context of this invention is understood to mean a monocyclic, bicyclic or polycyclic group.

(18) In the context of the present invention, a C.sub.1- to C.sub.20-alkyl group in which individual hydrogen atoms or CH.sub.2 groups may also be replaced by the abovementioned groups is understood to mean, for example, the methyl, ethyl, n-propyl, i-propyl, cyclopropyl, n-butyl, i-butyl, s-butyl, t-butyl, cyclobutyl, 2-methylbutyl, n-pentyl, s-pentyl, t-pentyl, 2-pentyl, neopentyl, cyclopentyl, n-hexyl, s-hexyl, t-hexyl, 2-hexyl, 3-hexyl, neoheptyl, cycloheptyl, 1-methylcyclopentyl, 2-methylpentyl, n-heptyl, 2-heptyl, 3-heptyl, 4-heptyl, cycloheptyl, 1-methylcyclohexyl, n-octyl, 2-ethylhexyl, cyclooctyl, 1-bicyclo[2.2.2]octyl, 2-bicyclo[2.2.2]octyl, 2-(2,6-dimethyl)octyl, 3-(3,7-dimethyl)octyl, adamantyl, trifluoromethyl, pentafluoroethyl, 2,2,2-

trifluoroethyl, 1,1-dimethyl-n-hex-1-yl, 1,1-dimethyl-n-hept-1-yl, 1,1-dimethyl-n-oct-1-yl, 1,1-dimethyl-n-dec-1-yl, 1,1-dimethyl-n-dodec-1-yl, 1,1-dimethyl-n-tetradec-1-yl, 1,1-dimethyl-n-hexadec-1-yl, 1,1-dimethyl-n-octadec-1-yl, 1,1-diethyl-n-hex-1-yl, 1,1-diethyl-n-hept-1-yl, 1,1-diethyl-n-oct-1-yl, 1,1-diethyl-n-dec-1-yl, 1,1-diethyl-n-dodec-1-yl, 1,1-diethyl-n-tetradec-1-yl, 1,1-diethyl-n-hexadec-1-yl, 1,1-diethyl-n-octadec-1-yl, 1-(n-propyl)cyclohex-1-yl, 1-(n-butyl)cyclohex-1-yl, 1-(n-hexyl)cyclohex-1-yl, 1-(n-octyl)cyclohex-1-yl and 1-(n-decyl)cyclohex-1-yl radicals. An alkenyl group is understood to mean, for example, ethenyl, propenyl, butenyl, pentenyl, cyclopentenyl, hexenyl, cyclohexenyl, heptenyl, cycloheptenyl, octenyl, cyclooctenyl or cyclooctadienyl. An alkynyl group is understood to mean, for example, ethynyl, propynyl, butynyl, pentynyl, hexynyl, heptynyl or octynyl. An OR.sup.1 group is understood to mean, for example, methoxy, trifluoromethoxy, ethoxy, n-propoxy, i-propoxy, n-butoxy, i-butoxy, s-butoxy, t-butoxy or 2-methylbutoxy.

(19) An aromatic or heteroaromatic ring system which has 5-40 aromatic ring atoms and may also be substituted in each case by the abovementioned radicals and which may be joined to the aromatic or heteroaromatic system via any desired positions is understood to mean, for example, groups derived from benzene, naphthalene, anthracene, benzanthracene, phenanthrene, benzophenanthrene, pyrene, chrysene, perylene, fluoranthene, benzoanthracene, naphthacene, pentacene, benzopyrene, biphenyl, biphenylene, terphenyl, terphenylene, fluorene, spirobifluorene, dihydrophenanthrene, dihydropyrene, tetrahydropyrene, cis- or trans-indenofluorene, cis- or trans-monobenzoindenofluorene, cis- or trans-dibenzoindenofluorene, truxene, isotruxene, spirotruxene, spiroisotruxene, furan, benzofuran, isobenzofuran, dibenzofuran, thiophene, benzothiophene, isobenzothiophene, dibenzothiophene, pyrrole, indole, isoindole, carbazole, indolocarbazole, indenocarbazole, pyridine, quinoline, isoquinoline, acridine, phenanthridine, benzo-5,6-quinoline, benzo-6,7-quinoline, benzo-7,8-quinoline, phenothiazine, phenoxazine, pyrazole, indazole, imidazole, benzimidazole, naphthimidazole, phenanthrimidazole, pyridimidazole, pyrazinimidazole, quinoxalinimidazole, oxazole, benzoxazole, naphthoxazole, anthroxazole, phenanthroxazole, isoxazole, 1,2-thiazole, 1,3-thiazole, benzothiazole, pyridazine, benzopyridazine, pyrimidine, benzopyrimidine, quinoxaline, 1,5-diazaanthracene, 2,7-diazapyrene, 2,3-diazapyrene, 1,6-diazapyrene, 1,8-diazapyrene, 4,5-diazapyrene, 4,5,9,10-tetraazaperylene, pyrazine, phenazine, phenoxazine, phenothiazine, fluorubine, naphthyridine, azacarbazole, benzocarboline, phenanthroline, 1,2,3-triazole, 1,2,4-triazole, benzotriazole, 1,2,3-oxadiazole, 1,2,4-oxadiazole, 1,2,5-oxadiazole, 1,3,4-oxadiazole, 1,2,3-thiadiazole, 1,2,4-thiadiazole, 1,2,5-thiadiazole, 1,3,4-thiadiazole, 1,3,5-triazine, 1,2,4-triazine, 1,2,3-triazine, tetrazole, 1,2,4,5-tetrazine, 1,2,3,4-tetrazine, 1,2,3,5-tetrazine, purine, pteridine, indolizine and benzothiadiazole.

(20) Stated hereinafter are preferred embodiments of the bridgehead V, i.e. the structure of the formula (3).

(21) Suitable embodiments of the group of the formula (3) are the structures of the following formulae (4) to (7):

(22) ##STR00007## where the symbols used have the definitions given above.

(23) In a preferred embodiment of the invention, all X.sup.1 groups in the group of the formula (3) are CR, and so the central trivalent cycle of the formula (3) is a benzene. In a further preferred embodiment of the invention, all X.sup.1 groups are a nitrogen atom, and so the central trivalent cycle of the formula (3) is a triazine. Preferred embodiments of the formula (3) are thus the structures of the formulae (4) and (5). Particular preference is given to the structure of the formula (4).

(24) The following is applicable in respect of preferred R radicals on the trivalent central benzene ring of the formula (4) or on the central pyrimidine ring of the formula (6) or on the central pyridine ring of the formula (7): R is the same or different at each instance and is H, D, F, CN, OR.sup.1, a straight-chain alkyl group having 1 to 10 carbon atoms or an alkenyl group having 2 to 10 carbon atoms or a branched or cyclic alkyl group having 3 to 10 carbon atoms, each of which

may be substituted by one or more R.sup.1 radicals but is preferably unsubstituted, or an aromatic or heteroaromatic ring system which has 5 to 24 aromatic ring atoms and may be substituted in each case by one or more R.sup.1 radicals; at the same time, the R radical may also form a ring system with an R radical on X.sup.2; R.sup.1 is the same or different at each instance and is H, D, F, CN, OR.sup.2, a straight-chain alkyl group having 1 to 10 carbon atoms or an alkenyl group having 2 to 10 carbon atoms or a branched or cyclic alkyl group having 3 to 10 carbon atoms, each of which may be substituted by one or more R.sup.2 radicals but is preferably unsubstituted, or an aromatic or heteroaromatic ring system which has 5 to 24 aromatic ring atoms and may be substituted in each case by one or more R.sup.2 radicals; at the same time, two or more adjacent R.sup.1 radicals together may form a ring system; R.sup.2 is the same or different at each instance and is H, D, F or an aliphatic, aromatic and/or heteroaromatic organic radical having 1 to 20 carbon atoms, in which one or more hydrogen atoms may also be replaced by F.

(25) The following is applicable in respect of particularly preferred R radicals on the trivalent central benzene ring of the formula (4) or on the central pyrimidine ring of the formula (6) or on the central pyridine ring of the formula (7): R is the same or different at each instance and is H, D, F, CN, a straight-chain alkyl group having 1 to 4 carbon atoms or a branched or cyclic alkyl group having 3 to 6 carbon atoms, each of which may be substituted by one or more R.sup.1 radicals but is preferably unsubstituted, or an aromatic or heteroaromatic ring system which has 6 to 12 aromatic ring atoms and may be substituted in each case by one or more R.sup.1 radicals; at the same time, the R radical may also form a ring system with an R radical on X.sup.2; R.sup.1 is the same or different at each instance and is H, D, F, CN, a straight-chain alkyl group having 1 to 4 carbon atoms or a branched or cyclic alkyl group having 3 to 6 carbon atoms, each of which may be substituted by one or more R.sup.2 radicals but is preferably unsubstituted, or an aromatic or heteroaromatic ring system which has 6 to 12 aromatic ring atoms and may be substituted in each case by one or more R.sup.2 radicals; at the same time, two or more adjacent R.sup.1 radicals together may form a ring system; R.sup.2 is the same or different at each instance and is H, D, F or an aliphatic or aromatic hydrocarbyl radical having 1 to 12 carbon atoms.

(26) Most preferably, the R radicals on the central benzene ring of the formula (4) or on the central pyrimidine ring of the formula (6) or on the central pyridine ring of the formula (7) are H. More preferably, the group of the formula (4) is a structure of the following formula (4'):

(27) ##STR00008## where the symbols used have the definitions given above.

(28) There follows a description of preferred bivalent arylene or heteroarylene units as occur in the group of the formulae (3) to (7). As apparent from structures of the formulae (3) to (7), these structures contain three ortho-bonded bivalent arylene or heteroarylene units.

(29) In a preferred embodiment of the invention, the symbol X.sup.3 is C, and so the group of the formula (3) can be represented by the group of the formula (3a) and the groups of the formulae (4) to (7) by the formulae (4a) to (7a):

(30) ##STR00009## where the symbols have the definitions listed above.

(31) The group of the formula (3) can be formally represented by the following formula (3'), where the formulae (3) and (3') encompass the same structures:

(32) ##STR00010## where Ar is the same or different in each case and is a group of the following formula (8):

(33) ##STR00011## where the dotted bond in each case represents the position of the bond of the bidentate sub-ligands L.sup.1, L.sup.2 or L.sup.3 to this structure, * represents the position of the linkage of the unit of the formula (8) to the central trivalent aryl or heteroaryl group and X.sup.2 and X.sup.3 have the definitions given above. Preferred substituents in the group of the formula (8) are selected from the above-described substituents R.

(34) The group of the formula (8) may represent a heteroaromatic five-membered ring or an aromatic or heteroaromatic six-membered ring. In a preferred embodiment of the invention, the group of the formula (8) contains not more than two heteroatoms in the aryl or heteroaryl group,

more preferably not more than one heteroatom. This does not mean that any substituents bonded to this group cannot also contain heteroatoms. In addition, this definition does not mean that formation of rings by substituents cannot give rise to fused aromatic or heteroaromatic structures, for example naphthalene, benzimidazole, etc. The group of the formula (8) is preferably selected from benzene, pyridine, pyrimidine, pyrazine, pyridazine, pyrrole, furan, thiophene, pyrazole, imidazole, oxazole and thiazole.

(35) When both X^{sup.3} groups in a cycle are carbon atoms, preferred embodiments of the group of the formula (8) are the structures of the following formulae (9) to (25):

(36) ##STR00012## ##STR00013## where the symbols used have the definitions given above.

(37) When one X^{sup.3} group in a cycle is a carbon atom and the other X^{sup.3} group in the same cycle is a nitrogen atom, preferred embodiments of the group of the formula (8) are the structures of the following formulae (26) to (33):

(38) ##STR00014## where the symbols used have the definitions given above.

(39) Particular preference is given to the optionally substituted six-membered aromatic rings and six-membered heteroaromatic rings of the formulae (9) to (13) depicted above. Very particular preference is given to ortho-phenylene, i.e. a group of the abovementioned formula (9), especially with R=H.

(40) At the same time, as described above in the description of the substituents, it is also possible for adjacent substituents together to form a ring system, such that fused structures, including fused aryl and heteroaryl groups, for example naphthalene, quinoline, benzimidazole, carbazole, dibenzofuran or dibenzothiophene, can form.

(41) The three groups of the formula (8) that are present in the group of the formulae (3) to (7) or formula (3') may be the same or different. In a preferred embodiment of the invention, all three groups in the formula (8) are the same and also have the same substitution.

(42) More preferably, the structures of the formula (4) to (7) are selected from the structures of the following formulae (4b) to (7b):

(43) ##STR00015## where the symbols used have the definitions given above.

(44) A preferred embodiment of the formula (4b) is the structure of the following formula (4b'):

(45) ##STR00016## where the symbols used have the definitions given above.

(46) More preferably, the R groups in the formulae (3) to (7) are the same or different at each instance and are H, D or an alkyl group having 1 to 4 carbon atoms. Most preferably, R=H. Very particular preference is thus given to the structures of the following formulae (4c) or (5c):

(47) ##STR00017## where the symbols used have the definitions given above.

(48) There follows a description of the bidentate sub-ligands L^{sup.1}. The sub-ligand L^{sup.1} is a monoanionic structure. This is indicated in formula (2) by a negative charge on the boron atom.

(49) In a preferred embodiment of the invention, all A groups are the same or different at each instance and are CR. The sub-ligand L^{sup.1} is thus preferably a structure of the following formula (2a):

(50) ##STR00018## where the symbols used have the definitions given above.

(51) The substituents R adjacent to the coordinating nitrogen atom identified by * are preferably selected from the group consisting of H, D, F, methyl, ethyl and phenyl, more preferably H and D, and most preferably H.

(52) More preferably, the sub-ligand L^{sup.1} thus has a structure of the following formula (2b):

(53) ##STR00019## where the symbols used have the definitions given above.

(54) The further R radicals in the structures of the formulae (2), (2a) and (2b) are preferably selected from the group consisting of H, D, F, Br, OR^{sup.1}, C(=O)R^{sup.1}, a straight-chain alkyl group having 1 to 10 carbon atoms or an alkenyl group having 2 to 10 carbon atoms or a branched or cyclic alkyl group having 3 to 10 carbon atoms, where the alkyl or alkenyl group may in each case be substituted by one or more R^{sup.1} radicals, or an aromatic or heteroaromatic ring system which has 5 to 24 aromatic ring atoms and may be substituted in each case by one or more R^{sup.1}

radicals; at the same time, two adjacent R radicals together may also form a ring system.

(55) More preferably, the R radicals in the structures of the formulae (2), (2a) and (2b) are selected from the group consisting of H, D, F, a straight-chain alkyl group having 1 to 5 carbon atoms or a branched or cyclic alkyl group which has 3 to 6 carbon atoms, where the alkyl group may be substituted by one or more R^{sup.1} radicals, but is preferably unsubstituted, or an aromatic or heteroaromatic ring system which has 6 to 13 aromatic ring atoms and may be substituted in each case by one or more R^{sup.1} radicals; at the same time, it is also possible for two adjacent R radicals together to form a ring system.

(56) Most preferably, the sub-ligand L^{sup.1} has a structure of the following formula (2c):

(57) ##STR00020## where the symbols used have the definitions given above.

(58) The substituents R^{sup.B} on the boron atom in formula (2), (2a), (2b) or (2c) are the same or different at each instance and are preferably selected from the group consisting of OR^{sup.1} where R^{sup.1} is an alkyl group having 1 to 5 carbon atoms, a straight-chain alkyl group having 1 to 5 carbon atoms, a branched or cyclic alkyl group having 3 to 6 carbon atoms, an aryl group which has 6 to 10 carbon atoms and may be substituted by one or more preferably nonaromatic R^{sup.1} radicals, or a heteroaryl group which has 5 to 10 aromatic ring atoms and may be substituted by one or more preferably nonaromatic R^{sup.1} radicals. The two R^{sup.B} radicals here may also together form a ring system.

(59) More preferably, the substituents R^{sup.B} on the boron atom in formula (2), (2a), (2b) or (2c) are the same or different at each instance and are selected from the group consisting of OR^{sup.1} where R^{sup.1} is methyl, ethyl, n-propyl or isopropyl, an alkyl group selected from methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, tert-butyl and neopentyl, especially methyl, a phenyl group which may be substituted by one or more alkyl groups having 1 to 5 carbon atoms, but is preferably unsubstituted, or a heteroaryl group having 5 or 6 aromatic ring atoms which may be substituted by one or more alkyl groups having 1 to 5 carbon atoms, but is preferably unsubstituted. The two R^{sup.B} radicals here may also together form a ring system.

(60) In a preferred embodiment of the invention, the two substituents R^{sup.B} are the same.

(61) When R^{sup.B} is a group of the formula OR^{sup.1}, it is preferable when the two substituents R^{sup.B} are OR^{sup.1} where R^{sup.1} in each case is an alkyl group having 1 to 5 carbon atoms and the two R^{sup.1} together form a ring system.

(62) When R^{sup.B} is a heteroaryl group, it is preferably a pyrazolyl group bonded to the boron atom via a nitrogen atom.

(63) When the two substituents R^{sup.B} together with the boron atom to which they are bonded form a ring system, preferred ring systems are the structures of the following formulae (B-1) to (B-8):

(64) ##STR00021## where R^{sup.1} and R^{sup.2} have the definitions given above, the dotted bonds to the boron atom each represent the bond to the pyrazolyl ring or, when A=N, the triazolyl ring, and the negative charge on the boron atom is not shown.

(65) In a preferred embodiment of the invention, L^{sup.3} is a sub-ligand of the formula (2), where the sub-ligands L^{sup.1} and L^{sup.3} may be the same or different. When L^{sup.3} is a sub-ligand of the formula (2), preferred embodiments for L^{sup.3} are the abovementioned preferred embodiments for L^{sup.1}. Where L^{sup.3} represents a sub-ligand of the formula (2), it is particularly preferable when L^{sup.1} and L^{sup.3} are the same and also have the same substitution. The compound of the formula (1) is thus preferably a compound of the following formula (1a):

(66) ##STR00022## where the symbols used have the definitions given above.

(67) There follows a description of the bidentate sub-ligand L^{sup.2} that coordinates to the iridium via one carbon atom and one nitrogen atom or via two carbon atoms. When L^{sup.3} is a bidentate sub-ligand that coordinates to the iridium via one carbon atom and one nitrogen atom or via two carbon atoms, preferred embodiments for this sub-ligand L^{sup.3} correspond to the preferred embodiments of L^{sup.2} set out above. In this case, it is particularly preferable when L^{sup.2} and

L.sup.3 are the same and also have the same substitution.

(68) Preferably, L.sup.2 has one carbon atom and one nitrogen atom as coordinating atoms.

(69) It is further preferable when the metallacycle which is formed from the iridium and the sub-ligand L.sup.2 is a five-membered ring. This is shown in schematic form below for C and N as coordinating atoms:

(70) ##STR00023## Five-membered ring where N is a coordinating nitrogen atom and C is a coordinating carbon atom, and the carbon atoms shown are atoms of the sub-ligand L.sup.2.

(71) In a preferred embodiment of the invention, the sub-ligand L.sup.2 is a structure of one of the following formulae (L-1) and (L-2):

(72) ##STR00024## where the dotted bond represents the bond of the sub-ligand to the V group, i.e. the group of the formula (3), and the other symbols used are as follows: CyC is the same or different at each instance and is a substituted or unsubstituted aryl or heteroaryl group which has 5 to 14 aromatic ring atoms and coordinates in each case to the metal via a carbon atom and which is bonded to CyD via a covalent bond; CyD is the same or different at each instance and is a substituted or unsubstituted heteroaryl group which has 5 to 14 aromatic ring atoms and coordinates to the metal via a nitrogen atom or via a carbene carbon atom and which is bonded to CyC via a covalent bond;

(73) at the same time, two or more of the optional substituents together may form a ring system. The optional radicals are preferably selected from the abovementioned R radicals.

(74) CyD preferably coordinates via an uncharged nitrogen atom or via a carbene carbon atom. In addition, CyC coordinates via an anionic carbon atom.

(75) When two or more of the substituents, especially two or more R radicals, together form a ring system, it is possible for a ring system to be formed from substituents bonded to directly adjacent carbon atoms. In addition, it is also possible that the substituents on CyC and CyD together form a ring, as a result of which CyC and CyD may also together form a single fused heteroaryl group as bidentate ligand.

(76) When the two sub-ligands L.sup.2 and L.sup.3 are sub-ligands that coordinate to the iridium via one carbon atom and one nitrogen atom or via two carbon atoms, the two sub-ligands L.sup.2 and L.sup.3 have a structure of the formula (L-1), or the two sub-ligands L.sup.2 and L.sup.3 have a structure of the formula (L-2), or one of the sub-ligands L.sup.2 and L.sup.3 has a structure of the formula (L-1) and the other of the sub-ligands has a structure of the formula (L-2). In a preferred embodiment of the invention, either both sub-ligands L.sup.2 and L.sup.3 have a structure of the formula (L-1) or both sub-ligands L.sup.2 and L.sup.3 have a structure of the formula (L-2).

(77) In a preferred embodiment of the present invention, CyC is an aryl or heteroaryl group having 6 to 13 aromatic ring atoms, more preferably having 6 to 10 aromatic ring atoms, most preferably having 6 aromatic ring atoms, which coordinates to the metal via a carbon atom, which may be substituted by one or more R radicals and which is bonded to CyD via a covalent bond.

(78) Preferred embodiments of the CyC group are the structures of the following formulae (CyC-1) to (CyC-19) where the CyC group binds in each case at the position signified by # to CyD and coordinates at the position signified by * to the iridium,

(79) ##STR00025## ##STR00026## ##STR00027## where R has the definitions given above and the other symbols used are as follows: X is the same or different at each instance and is CR or N, with the proviso that not more than two symbols X per cycle are N; W is the same or different at each instance and is NR, O or S;

(80) with the proviso that, when the group of the formula (3) is bonded to CyC, one symbol X is C and the group of the formula (3) is bonded to this carbon atom. When the CyC group is bonded to the group of the formula (3), the bond is preferably via the position marked by "o" in the formulae depicted above, and so the symbol X marked by "o" in that case is preferably C. The above-depicted structures which do not contain any symbol X marked by "o" are preferably not bonded directly to the group of the formula (3), since such a bond to the bridge is not advantageous for

steric reasons.

(81) Preferably, a total of not more than two symbols X in CyC are N, more preferably not more than one symbol X in CyC is N, and most preferably all symbols X are CR, with the proviso that, when the group of the formula (3) is bonded to CyC, one symbol X is C and the group of the formula (3) is bonded to this carbon atom.

(82) Particularly preferred CyC groups are the groups of the following formulae (CyC-1a) to (CyC-20a):

(83) ##STR00028## ##STR00029## ##STR00030## ##STR00031## where the symbols used have the definitions given above and, when the group of the formula (3) is bonded to CyC, one R radical is not present and the group of the formula (3) is bonded to the corresponding carbon atom. When the CyC group is bonded to the group of the formula (3), the bond is preferably via the position marked by "o" in the formulae depicted above, and so the R radical in this position in that case is preferably absent. The above-depicted structures which do not contain any carbon atom marked by "o" are preferably not bonded directly to the group of the formula (3).

(84) Preferred groups among the (CyC-1) to (CyC-19) groups are the (CyC-1), (CyC-3), (CyC-8), (CyC-10), (CyC-12), (CyC-13) and (CyC-16) groups, and particular preference is given to the (CyC-1a), (CyC-3a), (CyC-8a), (CyC-10a), (CyC-12a), (CyC-13a) and (CyC-16a) groups.

(85) In a further preferred embodiment of the invention, CyD is a heteroaryl group having 5 to 13 aromatic ring atoms, more preferably having 6 to 10 aromatic ring atoms, which coordinates to the metal via an uncharged nitrogen atom or via a carbene carbon atom and which may be substituted by one or more R radicals and which is bonded via a covalent bond to CyC.

(86) Preferred embodiments of the CyD group are the structures of the following formulae (CyD-1) to (CyD-12) where the CyD group binds in each case at the position signified by # to CyC and coordinates at the position signified by * to the iridium,

(87) ##STR00032## ##STR00033## where X, W and R have the definitions given above, with the proviso that, when the group of the formula (3) is bonded to CyD, one symbol X is C and the group of the formula (3) is bonded to this carbon atom. When the CyD group is bonded to the group of the formula (3), the bond is preferably via the position marked by "o" in the formulae depicted above, and so the symbol X marked by "o" in that case is preferably C. The above-depicted structures which do not contain any symbol X marked by "o" are preferably not bonded directly to the group of the formula (3), since such a bond to the bridge is not advantageous for steric reasons.

(88) In this case, the (CyD-1) to (CyD-4) and (CyD-7) to (CyD-12) groups coordinate to the metal via an uncharged nitrogen atom, and (CyD-5) and (CyD-6) groups via a carbene carbon atom.

(89) Preferably, a total of not more than two symbols X in CyD are N, more preferably not more than one symbol X in CyD is N, and especially preferably all symbols X are CR, with the proviso that, when the group of the formula (3) is bonded to CyD, one symbol X is C and the group of the formula (3) is bonded to this carbon atom.

(90) Particularly preferred CyD groups are the groups of the following formulae (CyD-1a) to (CyD-12b):

(91) ##STR00034## ##STR00035## where the symbols used have the definitions given above and, when the group of the formula (3) is bonded to CyD, one R radical is not present and the group of the formula (3) is bonded to the corresponding carbon atom. When the CyD group is bonded to the group of the formula (3), the bond is preferably via the position marked by "o" in the formulae depicted above, and so the R radical in this position in that case is preferably absent. The above-depicted structures which do not contain any carbon atom marked by "o" are preferably not bonded directly to the group of the formula (3).

(92) Preferred groups among the (CyD-1) to (CyD-12) groups are the (CyD-1), (CyD-2), (CyD-3), (CyD-4), (CyD-5) and (CyD-6) groups, especially (CyD-1), (CyD-2) and (CyD-3), and particular preference is given to the (CyD-1a), (CyD-2a), (CyD-3a), (CyD-4a), (CyD-5a) and (CyD-6a) groups, especially (CyD-1a), (CyD-2a) and (CyD-3a).

(93) In a preferred embodiment of the present invention, CyC is an aryl or heteroaryl group having 6 to 13 aromatic ring atoms, and at the same time CyD is a heteroaryl group having 5 to 13 aromatic ring atoms. More preferably, CyC is an aryl or heteroaryl group having 6 to 10 aromatic ring atoms, and at the same time CyD is a heteroaryl group having 5 to 10 aromatic ring atoms. Most preferably, CyC is an aryl or heteroaryl group having 6 aromatic ring atoms, and CyD is a heteroaryl group having 6 to 10 aromatic ring atoms. At the same time, CyC and CyD may be substituted by one or more R radicals.

(94) The abovementioned preferred (CyC-1) to (CyC-20) and (CyD-1) to (CyD-12) groups may be combined with one another as desired, provided that at least one of the CyC or CyD groups has a suitable attachment site to the group of the formula (3), suitable attachment sites being signified by "o" in the formulae given above.

(95) It is especially preferable when the CyC and CyD groups specified above as particularly preferred, i.e. the groups of the formulae (CyC-1a) to (CyC-20a) and the groups of the formulae (CyD1-a) to (CyD-14b), are combined with one another, provided that at least one of the preferred CyC or CyD groups has a suitable attachment site to the group of the formula (3), suitable attachment sites being signified by "o" in the formulae given above. Combinations in which neither CyC nor CyD has such a suitable attachment site for the group of the formula (3) are therefore not preferred.

(96) It is very particularly preferable when one of the (CyC-1), (CyC-3), (CyC-8), (CyC-10), (CyC-12), (CyC-13) and (CyC-16) groups and especially the (CyC-1a), (CyC-3a), (CyC-8a), (CyC-10a), (CyC-12a), (CyC-13a) and (CyC-16a) groups is combined with one of the (CyD-1), (CyD-2) and (CyD-3) groups and especially with one of the (CyD-1a), (CyD-2a) and (CyD-3a) groups.

(97) Preferred sub-ligands (L-1) are the structures of the formulae (L-1-1) and (L-1-2), and preferred sub-ligands (L-2) are the structures of the formulae (L-2-1) to (L-2-4):

(98) ##STR00036## where the symbols used have the definitions given above and "o" represents the position of the bond to the group of the formula (3).

(99) Particularly preferred sub-ligands (L-1) are the structures of the formulae (L-1-1a) and (L-1-2b), and particularly preferred sub-ligands (L-2) are the structures of the formulae (L-2-1a) to (L-2-4a)

(100) ##STR00037## ##STR00038## where the symbols used have the definitions given above and "o" represents the position of the bond to the group of the formula (3).

(101) When two R radicals of which one is bonded to CyC and the other to CyD together form an aromatic ring system, this can result in bridged sub-ligands and, for example, also in sub-ligands which overall constitute a single larger heteroaryl group, for example benzo[h]quinoline, etc. The ring between the substituents on CyC and CyD is preferably formed by a group of one of the following formulae (34) to (43):

(102) ##STR00039## where R.sup.1 has the definitions given above and the dotted bonds signify the bonds to CyC or CyD. At the same time, the unsymmetric groups among those mentioned above may be incorporated in each of the two possible options; for example, in the group of the formula (43), the oxygen atom may bind to the CyC group and the carbonyl group to the CyD group, or the oxygen atom may bind to the CyD group and the carbonyl group to the CyC group.

(103) Particular preference is given to the groups of the formulae (34) and (36). In addition, the group of the formula (40) is preferred especially when this results in ring formation to give a six-membered ring, as shown below, for example, by the formulae (L-21) and (L-22).

(104) Preferred ligands which arise through ring formation between two R radicals in the different cycles are the structures of the formulae (L-3) to (L-30) shown below:

(105) ##STR00040## ##STR00041## ##STR00042## ##STR00043## ##STR00044## where the symbols used have the definitions given above and "o" indicates the position at which this sub-ligand is joined to the group of the formula (3).

(106) In a preferred embodiment of the sub-ligands of the formulae (L-3) to (L-30), a total of one

symbol X is N and the other symbols X are CR, or all symbols X are CR.

(107) In a further embodiment of the invention, it is preferable if, in the groups (CyC-1) to (CyC-20) or (CyD-1) to (CyD-14) or in the sub-ligands (L-5) to (L-32), one of the atoms X is N when an R group bonded as a substituent adjacent to this nitrogen atom is not hydrogen or deuterium. This applies analogously to the preferred structures (CyC-1a) to (CyC-20a) or (CyD-1a) to (CyD-14b) in which a substituent bonded adjacent to a non-coordinating nitrogen atom is preferably an R group which is not hydrogen or deuterium.

(108) In this case, this substituent R is preferably a group selected from CF.sub.3, OCF.sub.3, alkyl groups having 1 to 10 carbon atoms, especially branched or cyclic alkyl groups having 3 to 10 carbon atoms, OR.sup.1 where R.sup.1 is an alkyl group having 1 to 10 carbon atoms, especially a branched or cyclic alkyl group having 3 to 10 carbon atoms, a dialkylamino group having 2 to 10 carbon atoms, aromatic or heteroaromatic ring systems or aralkyl or heteroaralkyl groups. These groups are sterically demanding groups. Further preferably, this R radical may also form a cycle with an adjacent R radical.

(109) A further suitable bidentate sub-ligand for metal complexes in which the metal is a transition metal is a sub-ligand of the following formula (L-31) or (L-32):

(110) ##STR00045## where R has the definitions given above, * represents the position of coordination to the metal, "o" represents the position of linkage of the sub-ligand to the group of the formula (3) and in addition: X is the same or different at each instance and is CR or N, with the proviso that not more than one X symbol per cycle is N.

(111) When two R radicals bonded to adjacent carbon atoms in the sub-ligands (L-31) and (L-32) form an aromatic cycle with one another, this cycle together with the two adjacent carbon atoms is preferably a structure of the formula (44):

(112) ##STR00046## where the dotted bonds symbolize the linkage of this group within the sub-ligand and Y is the same or different at each instance and is CR.sup.1 or N and preferably not more than one symbol Y is N.

(113) In a preferred embodiment of the sub-ligand (L-31) or (L-32), not more than one group of the formula (44) is present. The sub-ligands are thus preferably sub-ligands of the following formulae (L-33) to (L-38):

(114) ##STR00047## where X is the same or different at each instance and is CR or N, but the R radicals together do not form an aromatic or heteroaromatic ring system and the further symbols have the definitions given above.

(115) In a preferred embodiment of the invention, in the sub-ligand of the formulae (L-31) to (L-38), a total of 0, 1 or 2 of the symbols X and, if present, Y are N. More preferably, a total of 0 or 1 of the symbols X and, if present, Y are N.

(116) Preferred embodiments of the formulae (L-33) to (L-38) are the structures of the following formulae (L-33a) to (L-38f):

(117) ##STR00048## ##STR00049## ##STR00050## ##STR00051## ##STR00052##
##STR00053## ##STR00054## where the symbols used have the definitions given above and "o" indicates the position of the linkage to the group of the formula (3).

(118) In a preferred embodiment of the invention, the X group in the ortho position to the coordination to the metal is CR. In this radical, R bonded in the ortho position to the coordination to the metal is preferably selected from the group consisting of H, D, F and methyl.

(119) In a further embodiment of the invention, it is preferable, if one of the atoms X or, if present, Y is N, when a substituent bonded adjacent to this nitrogen atom is an R group which is not hydrogen or deuterium.

(120) In this case, this substituent R is preferably a group selected from CF.sub.3, OCF.sub.3, alkyl groups having 1 to 10 carbon atoms, especially branched or cyclic alkyl groups having 3 to 10 carbon atoms, OR.sup.1 where R.sup.1 is an alkyl group having 1 to 10 carbon atoms, especially a branched or cyclic alkyl group having 3 to 10 carbon atoms, a dialkylamino group having 2 to 10

carbon atoms, aromatic or heteroaromatic ring systems or aralkyl or heteroaralkyl groups. These groups are sterically demanding groups. Further preferably, this R radical may also form a cycle with an adjacent R radical.

(121) There follows a description of preferred substituents as may be present on the above-described sub-ligands L^{sup.1}, L^{sup.2} and L^{sup.3}, but also on the bivalent arylene or heteroarylene group in the structure of the formulae (3) to (7), i.e. in the structure of the formula (8).

(122) In one embodiment of the invention, the compound of the invention contains two substituents R which are bonded to adjacent carbon atoms and together form an aliphatic ring according to one of the formulae described hereinafter. In this case, the two substituents R which form this aliphatic ring may be present on the group of the formula (3) and/or on one or more of the bidentate sub-ligands. The aliphatic ring which is formed by the ring formation by two substituents R together is preferably described by one of the following formulae (45) to (51):

(123) ##STR00055## where R^{sup.1} and R^{sup.2} have the definitions given above, the dotted bonds signify the linkage of the two carbon atoms in the ligand and, in addition: A^{sup.1}, A^{sup.3} is the same or different at each instance and is C(R^{sup.3}).sub.2, O, S, NR^{sup.3} or C(=O); A^{sup.2} is C(R^{sup.1}).sub.2, O, S, NR^{sup.3} or C(=O); G is an alkylene group which has 1, 2 or 3 carbon atoms and may be substituted by one or more R^{sup.2} radicals, —CR^{sup.2}=CR^{sup.2}— or an ortho-bonded arylene or heteroarylene group which has 5 to 14 aromatic ring atoms and may be substituted by one or more R^{sup.2} radicals; R^{sup.3} is the same or different at each instance and is H, F, a straight-chain alkyl or alkoxy group having 1 to 10 carbon atoms, a branched or cyclic alkyl or alkoxy group having 3 to 10 carbon atoms, where the alkyl or alkoxy group may be substituted in each case by one or more R^{sup.2} radicals, where one or more nonadjacent CH_{sub.2} groups may be replaced by R^{sup.2}C=CR^{sup.2}, C≡C, Si(R^{sup.2}).sub.2, C=O, NR^{sup.2}, O, S or CONR^{sup.2}, or an aromatic or heteroaromatic ring system which has 5 to 24 aromatic ring atoms and may be substituted in each case by one or more R^{sup.2} radicals, or an aryloxy or heteroaryloxy group which has 5 to 24 aromatic ring atoms and may be substituted by one or more R^{sup.2} radicals; at the same time, two R^{sup.3} radicals bonded to the same carbon atom together may form an aliphatic or aromatic ring system and thus form a spiro system; in addition, R^{sup.3} with an adjacent R or R^{sup.1} radical may form an aliphatic ring system;

(124) with the proviso that no two heteroatoms in these groups are bonded directly to one another and no two C=O groups are bonded directly to one another.

(125) In the above-depicted structures of the formulae (45) to (51) and the further embodiments of these structures specified as preferred, a double bond is depicted in a formal sense between the two carbon atoms. This is a simplification of the chemical structure when these two carbon atoms are incorporated into an aromatic or heteroaromatic system and hence the bond between these two carbon atoms is formally between the bonding level of a single bond and that of a double bond. The drawing of the formal double bond should thus not be interpreted so as to limit the structure; instead, it will be apparent to the person skilled in the art that this is an aromatic bond.

(126) When adjacent radicals in the structures of the invention form an aliphatic ring system, it is preferable when the latter does not have any acidic benzylic protons. Benzylic protons are understood to mean protons which bind to a carbon atom bonded directly to the ligand. This can be achieved by virtue of the carbon atoms in the aliphatic ring system which bind directly to an aryl or heteroaryl group being fully substituted and not containing any bonded hydrogen atoms. Thus, the absence of acidic benzylic protons in the formulae (45) to (47) is achieved by virtue of A^{sup.1} and A^{sup.3}, when they are C(R^{sup.3}).sub.2, being defined such that R^{sup.3} is not hydrogen. This can additionally also be achieved by virtue of the carbon atoms in the aliphatic ring system which bind directly to an aryl or heteroaryl group being the bridgeheads in a bi- or polycyclic structure. The protons bonded to bridgehead carbon atoms, because of the spatial structure of the bi- or polycycle, are significantly less acidic than benzylic protons on carbon atoms which are not bonded within a

bi- or polycyclic structure, and are regarded as non-acidic protons in the context of the present invention.

(127) Thus, the absence of acidic benzylic protons in formulae (48) to (51) is achieved by virtue of this being a bicyclic structure, as a result of which R.sup.1, when it is H, is much less acidic than benzylic protons since the corresponding anion of the bicyclic structure is not mesomerically stabilized. Even when R.sup.1 in formulae (48) to (51) is H, this is therefore a non-acidic proton in the context of the present application.

(128) In a preferred embodiment of the invention, R.sup.3 is not H.

(129) In a preferred embodiment of the structure of the formulae (45) to (51), not more than one of the A.sup.1, A.sup.2 and A.sup.3 groups is a heteroatom, especially O or NR.sup.3, and the other groups are C(R.sup.3).sub.2 or C(R.sup.1).sub.2, or A.sup.1 and A.sup.3 are the same or different at each instance and are O or NR.sup.3 and A.sup.2 is C(R.sup.1).sub.2. In a particularly preferred embodiment of the invention, A.sup.1 and A.sup.3 are the same or different at each instance and are C(R.sup.3).sub.2, and A.sup.2 is C(R.sup.1).sub.2 and more preferably C(R.sup.3).sub.2 or CH.sub.2.

(130) Preferred embodiments of the formula (45) are thus the structures of the formulae (45-A), (45-B), (45-C) and (45-D), and a particularly preferred embodiment of the formula (45-A) is the structures of the formulae (45-E) and (45-F):

(131) ##STR00056## where R.sup.1 and R.sup.3 have the definitions given above and A.sup.1, A.sup.2 and A.sup.3 are the same or different at each instance and are O or NR.sup.3.

(132) Preferred embodiments of the formula (46) are the structures of the following formulae (46-A) to (46-F):

(133) ##STR00057## where R.sup.1 and R.sup.3 have the definitions given above and A.sup.1, A.sup.2 and A.sup.3 are the same or different at each instance and are O or NR.sup.3.

(134) Preferred embodiments of the formula (47) are the structures of the following formulae (47-A) to (47-E):

(135) ##STR00058## where R.sup.1 and R.sup.3 have the definitions given above and A.sup.1, A.sup.2 and A.sup.3 are the same or different at each instance and are O or NR.sup.3.

(136) In a preferred embodiment of the structure of formula (48), the R.sup.1 radicals bonded to the bridgehead are H, D, F or CH.sub.3. Further preferably, A.sup.2 is C(R.sup.1).sub.2 or O, and more preferably C(R.sup.3).sub.2. Preferred embodiments of the formula (48) are thus structures of the formulae (48-A) and (48-B), and a particularly preferred embodiment of the formula (48-A) is a structure of the formula (48-C):

(137) ##STR00059## where the symbols used have the definitions given above.

(138) In a preferred embodiment of the structure of formulae (49), (50) and (51), the R.sup.1 radicals bonded to the bridgehead are H, D, F or CH.sub.3. Further preferably, A.sup.2 is C(R.sup.1).sub.2. Preferred embodiments of the formulae (49), (50) and (51) are thus the structures of the formulae (49-A), (50-A) and (51-A):

(139) ##STR00060## where the symbols used have the definitions given above.

(140) Further preferably, the G group in the formulae (48), (48-A), (48-B), (48-C), (49), (49-A), (50), (50-A), (51) and (51-A) is a 1,2-ethylene group which may be substituted by one or more R.sup.2 radicals, where R.sup.2 is preferably the same or different at each instance and is H or an alkyl group having 1 to 4 carbon atoms, or an ortho-arylene group which has 6 to 10 carbon atoms and may be substituted by one or more R.sup.2 radicals, but is preferably unsubstituted, especially an ortho-phenylene group which may be substituted by one or more R.sup.2 radicals, but is preferably unsubstituted.

(141) In a further preferred embodiment of the invention, R.sup.3 in the groups of the formulae (45) to (51) and in the preferred embodiments is the same or different at each instance and is F, a straight-chain alkyl group having 1 to 10 carbon atoms or a branched or cyclic alkyl group having 3 to 20 carbon atoms, where one or more nonadjacent CH.sub.2 groups in each case may be replaced

by R.sup.2C=CR.sup.2 and one or more hydrogen atoms may be replaced by D or F, or an aromatic or heteroaromatic ring system which has 5 to 14 aromatic ring atoms and may be substituted in each case by one or more R.sup.2 radicals; at the same time, two R.sup.3 radicals bonded to the same carbon atom may together form an aliphatic or aromatic ring system and thus form a spiro system; in addition, R.sup.3 may form an aliphatic ring system with an adjacent R or R.sup.1 radical.

(142) In a particularly preferred embodiment of the invention, R.sup.3 in the groups of the formulae (45) to (51) and in the preferred embodiments is the same or different at each instance and is F, a straight-chain alkyl group having 1 to 3 carbon atoms, especially methyl, or an aromatic or heteroaromatic ring system which has 5 to 12 aromatic ring atoms and may be substituted in each case by one or more R.sup.2 radicals, but is preferably unsubstituted; at the same time, two R.sup.3 radicals bonded to the same carbon atom may together form an aliphatic or aromatic ring system and thus form a spiro system; in addition, R.sup.3 may form an aliphatic ring system with an adjacent R or R.sup.1 radical.

(143) Examples of particularly suitable groups of the formula (45) are the structures listed below:

(144) ##STR00061## ##STR00062## ##STR00063##

(145) Examples of particularly suitable groups of the formula (46) are the structures listed below:

(146) ##STR00064##

(147) Examples of particularly suitable groups of the formulae (47), (49) and (50) are the structures listed below:

(148) ##STR00065##

(149) Examples of particularly suitable groups of the formula (48) are the structures listed below:

(150) ##STR00066##

(151) Examples of particularly suitable groups of the formula (49) are the structures listed below:

(152) ##STR00067##

(153) In a further embodiment of the invention, the sub-ligand L.sup.2 is substituted by a substituent of the following formula (52) or (53), preferably in a position para to the coordination to the iridium:

(154) ##STR00068## where the dotted bond indicates the bond to the sub-ligand L.sup.2 and in addition: R' is the same or different at each instance and is selected from the group consisting of H, D, F, CN, a straight-chain alkyl group having 1 to 10 carbon atoms in which one or more hydrogen atoms may also be replaced by D or F, or a branched or cyclic alkyl group having 3 to 10 carbon atoms in which one or more hydrogen atoms may also be replaced by D or F, or an alkenyl group having 2 to 10 carbon atoms in which one or more hydrogen atoms may also be replaced by D or F; at the same time, two adjacent R' radicals or two R' radicals on adjacent phenyl groups together may also form a ring system; or two R' on adjacent phenyl groups together are a group selected from O and S, such that the two phenyl rings together with the bridging group are a dibenzofuran or dibenzothiophene, and the further R' are as defined above; n is 0, 1, 2, 3, 4 or 5.

(155) The presence of a substituent (52) or (53) can achieve oriented emission and hence a rise in efficiency.

(156) When the sub-ligand L.sup.2 has a substituent of the formula (52) or (53) and L.sup.3 is likewise a sub-ligand that coordinates to the iridium via one carbon atom and one nitrogen atom or two carbon atoms, it is preferable when this sub-ligand L.sup.3 has either no aromatic or heteroaromatic substituents or as substituents has solely aryl or heteroaryl groups that contain not more than 6 aromatic ring atoms and may be substituted solely by nonaromatic substituents.

(157) The groups of the formulae (52) and (53) differ merely in that the group of the formula (52) is bonded to the sub-ligand L.sup.2 in the para position and the group of the formula (53) in the meta position.

(158) In a preferred embodiment of the invention, in the formulae (52) and (53), n=0, 1 or 2, preferably 0 or 1 and most preferably 0.

(159) In a further preferred embodiment of the invention, both substituents R' bonded in the ortho positions to the carbon atom by which the group of the formula (52) or (53) is bonded to the phenylpyridine ligands are the same or different and are H or D.

(160) Preferred embodiments of the structure of the formula (52) are the structures of the formulae (52a) to (52n), and preferred embodiments of the structure of the formula (53) are the structures of the formulae (53a) to (53n):

(161) ##STR00069## ##STR00070## ##STR00071## ##STR00072## where the symbols used have the definitions given above and where the fluorene group in the 9 position may also be substituted by one or two alkyl groups each having 1 to 6 carbon atoms, preferably having 1 to 4 carbon atoms, more preferably by two methyl groups.

(162) Preferred substituents R' in the groups of the formula (52) or (53) or the preferred embodiments are selected from the group consisting of H, D, CN and an alkyl group having 1 to 4 carbon atoms, more preferably H, D or methyl.

(163) The substituents of the formulae (52) and (53) are preferably used in complexes in which the sub-ligand L^{sup.3} is the same as or different from the sub-ligand L^{sup.1}, i.e. in complexes that have two sub-ligands of the formula (2).

(164) When R radicals are incorporated in the bidentate sub-ligands or in the bivalent arylene or heteroarylene groups of the formula (8) bonded within the formulae (3) to (7) or the preferred embodiments, these R radicals are the same or different at each instance and are preferably selected from the group consisting of H, D, F, Br, I, N(R^{sup.1}).sub.2, CN, Si(R^{sup.1}).sub.3, B(OR^{sup.1}).sub.2, C(=O)R^{sup.1}, a straight-chain alkyl group having 1 to 10 carbon atoms or an alkenyl group having 2 to 10 carbon atoms or a branched or cyclic alkyl group having 3 to 10 carbon atoms, where the alkyl or alkenyl group may be substituted in each case by one or more R^{sup.1} radicals, or an aromatic or heteroaromatic ring system which has 5 to 30 aromatic ring atoms and may be substituted in each case by one or more R^{sup.1} radicals; at the same time, two adjacent R radicals together or R together with R^{sup.1} may also form a mono- or polycyclic, aliphatic or aromatic ring system. More preferably, these R radicals are the same or different at each instance and are selected from the group consisting of H, D, F, N(R^{sup.1}).sub.2, a straight-chain alkyl group having 1 to 6 carbon atoms or a branched or cyclic alkyl group having 3 to 10 carbon atoms, where one or more hydrogen atoms may be replaced by D or F, or an aromatic or heteroaromatic ring system which has 5 to 24 aromatic ring atoms, especially 6 to 13 aromatic ring atoms, and may be substituted in each case by one or more R^{sup.1} radicals; at the same time, two adjacent R radicals together or R together with R^{sup.1} may also form a mono- or polycyclic, aliphatic or aromatic ring system.

(165) Preferred R^{sup.1} radicals are the same or different at each instance and are H, D, F, N(R^{sup.2}).sub.2, CN, a straight-chain alkyl group having 1 to 10 carbon atoms or an alkenyl group having 2 to 10 carbon atoms or a branched or cyclic alkyl group having 3 to 10 carbon atoms, where the alkyl group may be substituted in each case by one or more R^{sup.2} radicals, or an aromatic or heteroaromatic ring system which has 5 to 24 aromatic ring atoms and may be substituted in each case by one or more R^{sup.2} radicals; at the same time, two or more adjacent R^{sup.1} radicals together may form a mono- or polycyclic aliphatic ring system. Particularly preferred R^{sup.1} radicals bonded to R are the same or different at each instance and are H, F, CN, a straight-chain alkyl group having 1 to 5 carbon atoms or a branched or cyclic alkyl group having 3 to 5 carbon atoms, each of which may be substituted by one or more R^{sup.2} radicals, or an aromatic or heteroaromatic ring system which has 5 to 13 aromatic ring atoms and may be substituted in each case by one or more R^{sup.2} radicals; at the same time, two or more adjacent R^{sup.1} radicals together may form a mono- or polycyclic aliphatic ring system.

(166) Preferred R^{sup.2} radicals are the same or different at each instance and are H, F or an aliphatic hydrocarbyl radical having 1 to 5 carbon atoms or an aromatic hydrocarbyl radical having 6 to 12 carbon atoms; at the same time, two or more R^{sup.2} substituents together may also form a

mono- or polycyclic aliphatic ring system.

(167) The abovementioned preferred embodiments can be combined with one another as desired. In a particularly preferred embodiment of the invention, the abovementioned preferred embodiments apply simultaneously. Thus, it is especially preferable when the preferred embodiments of the V group are combined with the preferred embodiments of the sub-ligands L.sup.1, L.sup.2 and L.sup.3 and the preferred embodiments of the substituents.

(168) Examples of suitable structures of the invention are the compounds depicted below.

(169) The iridium complexes of the invention are chiral structures. If the tripodal ligand of the complexes is additionally also chiral, the formation of diastereomers and multiple enantiomer pairs is possible. In that case, the

(170) ##STR00073## ##STR00074## ##STR00075## ##STR00076## ##STR00077##

(171) complexes of the invention include both the mixtures of the different diastereomers or the corresponding racemates and the individual isolated diastereomers or enantiomers.

(172) If C.sub.s-symmetric ligand precursors having two identical precursors for sub-ligands L.sup.1 or two identical sub-ligands L.sup.2 are used in the complexation reaction, what is typically obtained is a racemic mixture of the C.sub.1-

(173) ##STR00078## ##STR00079## ##STR00080##

(174) symmetric complexes, i.e. of the A and the A enantiomer. These may be separated by standard methods (chromatography on chiral materials/columns or optical resolution by crystallization). This is shown in the scheme which follows using the example of a C.sub.s-symmetric ligand

(175) ##STR00081## ##STR00082##

(176) precursor bearing two identical phenylpyridine sub-ligands and also applies analogously to all other C.sub.s-symmetric ligand precursors.

(177) ##STR00083## ##STR00084## ##STR00085## ##STR00086## ##STR00087##

##STR00088## ##STR00089## ##STR00090## ##STR00091## ##STR00092## ##STR00093##
##STR00094## ##STR00095## ##STR00096## ##STR00097## ##STR00098## ##STR00099##
##STR00100## ##STR00101## ##STR00102## ##STR00103## ##STR00104## ##STR00105##
##STR00106## ##STR00107## ##STR00108## ##STR00109## ##STR00110## ##STR00111##
##STR00112## ##STR00113## ##STR00114## ##STR00115## ##STR00116## ##STR00117##
##STR00118## ##STR00119## ##STR00120## ##STR00121## ##STR00122## ##STR00123##
##STR00124## ##STR00125## ##STR00126## ##STR00127## ##STR00128## ##STR00129##
##STR00130## ##STR00131## ##STR00132## ##STR00133## ##STR00134## ##STR00135##
##STR00136## ##STR00137## ##STR00138## ##STR00139## ##STR00140## ##STR00141##
##STR00142## ##STR00143##

(178) ##STR00144##

(179) Optical resolution via fractional crystallization of diastereomeric salt pairs can be effected by customary methods. One option for this purpose is to oxidize the uncharged Ir(III) complexes (for example with peroxides or H.sub.2O.sub.2 or by electrochemical means), add the salt of an enantiomerically pure monoanionic base (chiral base) to the cationic Ir(IV) complexes thus produced, separate the diastereomeric salts thus produced by fractional crystallization, and then reduce them with the aid of a reducing agent (e.g. zinc, hydrazine hydrate, ascorbic acid, etc.) to give the enantiomerically pure uncharged complex, as shown schematically below:

(180) ##STR00145##

(181) In addition, an enantiomerically pure or enantiomerically enriching synthesis is possible by complexation in a chiral medium (e.g. R- or S-1,1-binaphthol).

(182) Analogous processes can also be conducted with complexes of C.sub.1-symmetric ligand precursors.

(183) If C.sub.1-symmetric ligands are used in the complexation, what is typically obtained is a diastereomer mixture of the complexes which can be separated by standard methods

(chromatography, crystallization).

(184) The compounds of the invention are preparable in principle by various processes. Generally, for this purpose, an iridium salt is reacted with an appropriate ligand precursor in the presence of a halogen scavenger, for example a silver salt (AgNO_3 , Ag_2O , Ag_2CO_3 , AgOTf etc.), and in the presence of the desired pyrazolylborate synthon. Suitable pyrazolylborate synthons may be uncharged or anionic. In the course of the reaction, as well as the o-metallation reaction, a coordination reaction and the template-controlled construction of the bis(pyrazolylborato) sub-ligand take place.

(185) ##STR00146##

(186) Therefore, the present invention further provides a process for preparing the compounds of the invention by reacting the appropriate ligand precursor with a pyrazolylborate synthon and with iridium alkoxides of the formula (54), with iridium ketoketonates of the formula (55), with iridium halides of the formula (56), and the trisacetone nitrile, trisbenzonitrile or tristetrahydrothiophene adducts thereof, or with iridium carboxylates of the formula (57),

(187) ##STR00147## where R has the definitions given above, $\text{Hal} = \text{F}$, Cl , Br or I and the iridium reactants may also take the form of the corresponding hydrates. R here is preferably an alkyl group having 1 to 4 carbon atoms.

(188) It is likewise possible to use iridium compounds bearing both alkoxide and/or halide and/or hydroxyl and ketoketonate radicals. These compounds may also be charged. Corresponding iridium compounds of particular suitability as reactants are disclosed in WO 2004/085449. Particularly suitable are $[\text{IrCl}_2(\text{acac})_2]^-$, for example $\text{Na}[\text{IrCl}_2(\text{acac})_2]$, metal complexes with acetylacetonate derivatives as ligand, for example $\text{Ir}(\text{acac})_3$ or tris(2,2,6,6-tetramethylheptane-3,5-dionato)iridium, and $\text{IrCl}_3 \cdot x\text{H}_2\text{O}$ where x is typically a number from 2 to 4.

(189) If appropriate, it is possible to add further additives such as acids or bases.

(190) It is possible by these processes, if necessary followed by purification, for example recrystallization or sublimation, to obtain the inventive compounds of formula (1) in high purity, preferably more than 99% (determined by means of ^1H NMR and/or HPLC).

(191) The compounds of the invention may be rendered soluble by suitable substitution, for example by comparatively long alkyl groups (about 4 to 20 carbon atoms), especially branched alkyl groups, or optionally substituted aryl groups, for example xylyl, mesityl or branched terphenyl or quaterphenyl groups. Another particular method that leads to a distinct improvement in the solubility of the metal complexes is the use of fused-on aliphatic groups, as shown, for example, by the formulae (45) to (51) disclosed above. Such compounds are then soluble in sufficient concentration at room temperature in standard organic solvents, for example toluene or xylene, to be able to process the complexes from solution. These soluble compounds are of particularly good suitability for processing from solution, for example by printing methods.

(192) For the processing of the iridium complexes of the invention from a liquid phase, for example by spin-coating or by printing methods, formulations of the iridium complexes of the invention are required. These formulations may, for example, be solutions, dispersions or emulsions. For this purpose, it may be preferable to use mixtures of two or more solvents.

(193) Suitable and preferred solvents are, for example, toluene, anisole, o-, m- or p-xylene, methyl benzoate, mesitylene, tetralin, veratrole, THF, methyl-THF, THP, chlorobenzene, dioxane, phenoxytoluene, especially 3-phenoxytoluene, (-)-fenchone, 1,2,3,5-tetramethylbenzene, 1,2,4,5-tetramethylbenzene, 1-methylnaphthalene, 2-methylbenzothiazole, 2-phenoxyethanol, 2-pyrrolidinone, 3-methylanisole, 4-methylanisole, 3,4-dimethylanisole, 3,5-dimethylanisole, acetophenone, α -terpineol, benzothiazole, butyl benzoate, cumene, cyclohexanol, cyclohexanone, cyclohexylbenzene, decalin, dodecylbenzene, ethyl benzoate, indane, NMP, p-cymene, phenetole, 1,4-diisopropylbenzene, dibenzyl ether, diethylene glycol butyl methyl ether, triethylene glycol butyl methyl ether, diethylene glycol dibutyl ether, triethylene glycol dimethyl ether, diethylene

glycol monobutyl ether, tripropylene glycol dimethyl ether, tetraethylene glycol dimethyl ether, 2-isopropyl-naphthalene, pentylbenzene, hexylbenzene, heptylbenzene, octylbenzene, 1,1-bis(3,4-dimethylphenyl)ethane, hexamethylindane, 2-methylbiphenyl, 3-methylbiphenyl, 1-methylnaphthalene, 1-ethylnaphthalene, ethyl octanoate, diethyl sebacate, octyl octanoate, heptylbenzene, menthyl isovalerate, cyclohexyl hexanoate or mixtures of these solvents.

(194) The present invention therefore further provides a formulation comprising at least one compound of the invention and at least one further compound. The further compound may, for example, be a solvent, especially one of the abovementioned solvents or a mixture of these solvents. The further compound may alternatively be a further organic or inorganic compound which is likewise used in the electronic device, for example a matrix material. This further compound may also be polymeric.

(195) The compound of the invention can be used in the electronic device as active component, preferably as emitter in the emissive layer or as hole or electron transport material in a hole- or electron-transporting layer, or as oxygen sensitizers or as photoinitiator or photocatalyst. The present invention thus further provides for the use of a compound of the invention in an electronic device or as oxygen sensitizer or as photoinitiator or photocatalyst. Enantiomerically pure iridium complexes of the invention are suitable as photocatalysts for chiral photoinduced syntheses.

(196) The present invention still further provides an electronic device comprising at least one compound of the invention.

(197) An electronic device is understood to mean any device comprising anode, cathode and at least one layer, said layer comprising at least one organic or organometallic compound. The electronic device of the invention thus comprises anode, cathode and at least one layer containing at least one iridium complex of the invention. Preferred electronic devices are selected from the group consisting of organic electroluminescent devices (OLEDs, PLEDs), organic integrated circuits (O-ICs), organic field-effect transistors (O-FETs), organic thin-film transistors (O-TFTs), organic light-emitting transistors (O-LETs), organic solar cells (O-SCs), the latter being understood to mean both purely organic solar cells and dye-sensitized solar cells, organic optical detectors, organic photoreceptors, organic field-quench devices (O-FQDs), light-emitting electrochemical cells (LECs), oxygen sensors and organic laser diodes (O-lasers), comprising at least one compound of the invention in at least one layer. Compounds that emit in the infrared are suitable for use in organic infrared electroluminescent devices and infrared sensors. Particular preference is given to organic electroluminescent devices. Active components are generally the organic or inorganic materials introduced between the anode and cathode, for example charge injection, charge transport or charge blocker materials, but especially emission materials and matrix materials. The compounds of the invention exhibit particularly good properties as emission material in organic electroluminescent devices. A preferred embodiment of the invention is therefore organic electroluminescent devices. In addition, the compounds of the invention can be used for production of singlet oxygen or in photocatalysis.

(198) The organic electroluminescent device comprises cathode, anode and at least one emitting layer. Apart from these layers, it may comprise still further layers, for example in each case one or more hole injection layers, hole transport layers, hole blocker layers, electron transport layers, electron injection layers, exciton blocker layers, electron blocker layers, charge generation layers and/or organic or inorganic p/n junctions. In this case, it is possible that one or more hole transport layers are p-doped, for example with metal oxides such as MoO₃ or WO₃, or with (per)fluorinated electron-deficient aromatics or with electron-deficient cyano-substituted heteroaromatics (for example according to JP 4747558, JP 2006-135145, US 2006/0289882, WO 2012/095143), or with quinoid systems (for example according to EP1336208) or with Lewis acids, or with boranes (for example according to US 2003/0006411, WO 2002/051850, WO 2015/049030) or with carboxylates of the elements of main group 3, 4 or 5 (WO 2015/018539), and/or that one or more electron transport layers are n-doped.

(199) It is likewise possible for interlayers to be introduced between two emitting layers, which have, for example, an exciton-blocking function and/or control charge balance in the electroluminescent device and/or generate charges (charge generation layer, for example in layer systems having two or more emitting layers, for example in white-emitting OLED components). However, it should be pointed out that not necessarily every one of these layers need be present.

(200) In this case, it is possible for the organic electroluminescent device to contain an emitting layer, or for it to contain a plurality of emitting layers. If a plurality of emission layers are present, these preferably have several emission maxima between 380 nm and 750 nm overall, such that the overall result is white emission; in other words, various emitting compounds which may fluoresce or phosphoresce are used in the emitting layers. Especially preferred are three-layer systems where the three layers exhibit blue, green and orange or red emission (for the basic construction see, for example, WO 2005/011013), or systems having more than three emitting layers. The system may also be a hybrid system wherein one or more layers fluoresce and one or more other layers phosphoresce. A preferred embodiment is tandem OLEDs. White-emitting organic electroluminescent devices may be used for lighting applications or else with colour filters for full-colour displays.

(201) In a preferred embodiment of the invention, the organic electroluminescent device comprises the iridium complex of the invention as emitting compound in one or more emitting layers.

(202) When the iridium complex of the invention is used as emitting compound in an emitting layer, it is preferably used in combination with one or more matrix materials.

(203) When the organic electroluminescent device is produced by vacuum vapour deposition, the mixture of the iridium complex of the invention and the matrix material contains between 0.1% and 99% by volume, preferably between 1% and 90% by volume, more preferably between 3% and 40% by volume and especially between 5% and 15% by volume of the iridium complex of the invention, based on the overall mixture of emitter and matrix material. Correspondingly, the mixture contains between 99.9% and 1% by volume, preferably between 99% and 10% by volume, more preferably between 97% and 60% by volume and especially between 95% and 85% by volume of the matrix material, based on the overall mixture of emitter and matrix material.

(204) When the organic electroluminescent device is produced from solution, the mixture of the iridium complex of the invention and the matrix material contains between 0.1% and 99% by weight, preferably between 1% and 90% by weight, more preferably between 3% and 30% by weight and especially between 3% and 20% by weight of the iridium complex of the invention, based on the overall mixture of emitter and matrix material. Correspondingly, the mixture contains between 99.9% and 1% by weight, preferably between 99% and 10% by weight, more preferably between 97% and 70% by weight and especially between 97% and 80% by weight of the matrix material, based on the overall mixture of emitter and matrix material.

(205) The matrix material used may generally be any materials which are known for the purpose according to the prior art. The triplet level of the matrix material is preferably higher than the triplet level of the emitter.

(206) Suitable matrix materials for the compounds of the invention are ketones, phosphine oxides, sulfoxides and sulfones, for example according to WO 2004/013080, WO 2004/093207, WO 2006/005627 or WO 2010/006680, triaryl amines, carbazole derivatives, e.g. CBP (N,N-biscarbazolylbiphenyl), m-CBP or the carbazole derivatives disclosed in WO 2005/039246, US 2005/0069729, JP 2004/288381, EP 1205527, WO 2008/086851 or US 2009/0134784, biscarbazole derivatives, indolocarbazole derivatives, for example according to WO 2007/063754 or WO 2008/056746, indenocarbazole derivatives, for example according to WO 2010/136109 or WO 2011/000455, azacarbazoles, for example according to EP 1617710, EP 1617711, EP 1731584, JP 2005/347160, bipolar matrix materials, for example according to WO 2007/137725, silanes, for example according to WO 2005/111172, azaboroles or boronic esters, for example according to WO 2006/117052, diazasilole derivatives, for example according to WO 2010/054729,

diazaphosphole derivatives, for example according to WO 2010/054730, triazine derivatives, for example according to WO 2010/015306, WO 2007/063754 or WO 2008/056746, zinc complexes, for example according to EP 652273 or WO 2009/062578, dibenzofuran derivatives, for example according to WO 2009/148015 or WO 2015/169412, or bridged carbazole derivatives, for example according to US 2009/0136779, WO 2010/050778, WO 2011/042107 or WO 2011/088877. Suitable matrix materials for solution-processed OLEDs are also polymers, for example according to WO 2012/008550 or WO 2012/048778, oligomers or dendrimers, for example according to Journal of Luminescence 183 (2017), 150-158.

(207) It may also be preferable to use a plurality of different matrix materials as a mixture, especially at least one electron-conducting matrix material and at least one hole-conducting matrix material. A preferred combination is, for example, the use of an aromatic ketone, a triazine derivative or a phosphine oxide derivative with a triarylamine derivative or a carbazole derivative as mixed matrix for the metal complex of the invention.

(208) Preference is likewise given to the use of a mixture of a charge-transporting matrix material and an electrically inert matrix material (called a "wide bandgap host") having no significant involvement, if any, in the charge transport, as described, for example, in WO 2010/108579 or WO 2016/184540. Preference is likewise given to the use of two electron-transporting matrix materials, for example triazine derivatives and lactam derivatives, as described, for example, in WO 2014/094964.

(209) Depicted below are examples of compounds that are suitable as matrix materials for the compounds of the invention.

(210) Examples of triazines and pyrimidines which can be used as electron-transporting matrix materials are the following structures:

(211) ##STR00148## ##STR00149## ##STR00150## ##STR00151## ##STR00152##
##STR00153## ##STR00154## ##STR00155## ##STR00156## ##STR00157## ##STR00158##
##STR00159## ##STR00160## ##STR00161## ##STR00162## ##STR00163## ##STR00164##
##STR00165## ##STR00166## ##STR00167## ##STR00168## ##STR00169## ##STR00170##
##STR00171## ##STR00172## ##STR00173## ##STR00174## ##STR00175## ##STR00176##
##STR00177## ##STR00178## ##STR00179## ##STR00180## ##STR00181## ##STR00182##
##STR00183## ##STR00184## ##STR00185## ##STR00186## ##STR00187## ##STR00188##
##STR00189##

(212) Examples of lactams which can be used as electron-transporting matrix materials are the following structures:

(213) ##STR00190## ##STR00191## ##STR00192## ##STR00193## ##STR00194##
##STR00195## ##STR00196## ##STR00197## ##STR00198##

(214) Examples of indolo- and indenocarbazole derivatives in the broadest sense which can be used as hole- or electron-transporting matrix materials according to the substitution pattern are the following structures:

(215) ##STR00199## ##STR00200## ##STR00201## ##STR00202## ##STR00203##
##STR00204## ##STR00205## ##STR00206## ##STR00207## ##STR00208## ##STR00209##
##STR00210## ##STR00211## ##STR00212## ##STR00213## ##STR00214## ##STR00215##
##STR00216## ##STR00217## ##STR00218## ##STR00219## ##STR00220## ##STR00221##
##STR00222## ##STR00223## ##STR00224## ##STR00225## ##STR00226## ##STR00227##
##STR00228## ##STR00229## ##STR00230## ##STR00231## ##STR00232## ##STR00233##
##STR00234## ##STR00235##

(216) Examples of carbazole derivatives which can be used as hole- or electron-transporting matrix materials according to the substitution pattern are the following structures:

(217) ##STR00236## ##STR00237## ##STR00238## ##STR00239## ##STR00240##
##STR00241## ##STR00242## ##STR00243##

(218) Examples of bridged carbazole derivatives which can be used as hole-transporting matrix

materials are the following structures:

(219) ##STR00244## ##STR00245## ##STR00246## ##STR00247## ##STR00248##
##STR00249## ##STR00250## ##STR00251## ##STR00252## ##STR00253## ##STR00254##
##STR00255## ##STR00256## ##STR00257## ##STR00258## ##STR00259## ##STR00260##
##STR00261##

(220) Examples of biscarbazole derivatives which can be used as hole-transporting matrix materials are the following structures:

(221) ##STR00262## ##STR00263## ##STR00264## ##STR00265## ##STR00266##
##STR00267## ##STR00268## ##STR00269## ##STR00270## ##STR00271## ##STR00272##
##STR00273## ##STR00274## ##STR00275## ##STR00276## ##STR00277## ##STR00278##
##STR00279## ##STR00280## ##STR00281## ##STR00282## ##STR00283## ##STR00284##

(222) Examples of amines which can be used as hole-transporting matrix materials are the following structures:

(223) ##STR00285## ##STR00286## ##STR00287## ##STR00288## ##STR00289##
##STR00290## ##STR00291## ##STR00292## ##STR00293## ##STR00294## ##STR00295##
##STR00296## ##STR00297## ##STR00298## ##STR00299## ##STR00300##

(224) Examples of materials which can be used as wide bandgap matrix materials are the following structures:

(225) ##STR00301## ##STR00302## ##STR00303##

(226) In a further embodiment of the invention, a mixture of two or more triplet emitters, especially two or three triplet emitters, together with one or more matrix materials is used. In this case, the triplet emitter having the shorter-wave emission spectrum serves as co-matrix for the triplet emitter having the longer-wave emission spectrum. These triplet emitters preferably have the emission colours of green and orange or red, or alternatively blue and green. For example, the metal complexes of the invention can be combined with a metal complex emitting at shorter wavelength as co-matrix. For example, it is also possible to use metal complexes of the invention as co-matrix for triplet emitters that emit at longer wavelength. In this case, it may also be preferable when both the shorter-wave- and the longer-wave-emitting metal complex is a compound of the invention. A preferred embodiment in the case of use of a mixture of three triplet emitters is when two are used as co-host and one as emitting material. These triplet emitters preferably have the emission colours of green, yellow and red, or alternatively blue, green and orange.

(227) A further preferred mixture comprises, in the emitting layer, a charge-transporting host material, especially an electron-transporting host material, and what is called a “wide bandgap” host material which, owing to its electronic properties, is not involved to a significant degree, if at all, in the charge transport in the layer, and a co-dopant which is a triplet emitter which emits at a shorter wavelength than the compound of the invention, and a compound of the invention.

(228) The compounds of the invention can also be used in other functions in the electronic device, for example as hole transport material in a hole injection or transport layer, as charge generation material, as electron blocker material, as hole blocker material or as electron transport material, for example in an electron transport layer. It is likewise possible to use the compounds of the invention as matrix material for other phosphorescent metal complexes in an emitting layer.

(229) Preferred cathodes are metals having a low work function, metal alloys or multilayer structures composed of various metals, for example alkaline earth metals, alkali metals, main group metals or lanthanoids (e.g. Ca, Ba, Mg, Al, In, Mg, Yb, Sm, etc.). Additionally suitable are alloys composed of an alkali metal or alkaline earth metal and silver, for example an alloy composed of magnesium and silver. In the case of multilayer structures, in addition to the metals mentioned, it is also possible to use further metals having a relatively high work function, for example Ag, in which case combinations of the metals such as Mg/Ag, Ca/Ag or Ba/Ag, for example, are generally used. It may also be preferable to introduce a thin interlayer of a material having a high dielectric constant between a metallic cathode and the organic semiconductor. Examples of useful materials

for this purpose are alkali metal or alkaline earth metal fluorides, but also the corresponding oxides or carbonates (e.g. LiF, Li_2O , BaF_2 , MgO , NaF , CsF , Cs_2CO_3 , etc.). Likewise useful for this purpose are organic alkali metal complexes, e.g. Liq (lithium quinolate). The layer thickness of this layer is preferably between 0.5 and 5 nm.

(230) Preferred anodes are materials having a high work function. Preferably, the anode has a work function of greater than 4.5 eV versus vacuum. Firstly, metals having a high redox potential are suitable for this purpose, for example Ag, Pt or Au. Secondly, metal/metal oxide electrodes (e.g. Al/Ni/NiO_x , Al/PtO_x) may also be preferred. For some applications, at least one of the electrodes has to be transparent or partly transparent in order to enable either the irradiation of the organic material (O-SC) or the emission of light (OLED/PLED, O-laser). Preferred anode materials here are conductive mixed metal oxides. Particular preference is given to indium tin oxide (ITO) or indium zinc oxide (IZO). Preference is further given to conductive doped organic materials, especially conductive doped polymers, for example PEDOT, PANI or derivatives of these polymers. It is further preferable when a p-doped hole transport material is applied to the anode as hole injection layer, in which case suitable p-dopants are metal oxides, for example MoO_3 or WO_3 , or (per)fluorinated electron-deficient aromatic systems. Further suitable p-dopants are HAT-CN (hexacyanoheptaazatriphenylene) or the compound NPD9 from Novaled. Such a layer simplifies hole injection into materials having a low HOMO, i.e. a large HOMO in terms of magnitude.

(231) In the further layers, it is generally possible to use any materials as used according to the prior art for the layers, and the person skilled in the art is able, without exercising inventive skill, to combine any of these materials with the materials of the invention in an electronic device.

(232) Suitable charge transport materials as usable in the hole injection or hole transport layer or electron blocker layer or in the electron transport layer of the organic electroluminescent device of the invention are, for example, the compounds disclosed in Y. Shirota et al., Chem. Rev. 2007, 107(4), 953-1010, or other materials as used in these layers according to the prior art. Preferred hole transport materials which can be used in a hole transport, hole injection or electron blocker layer in the electroluminescent device of the invention are indenofluorenamine derivatives (for example according to WO 06/122630 or WO 06/100896), the amine derivatives disclosed in EP 1661888, hexaazatriphenylene derivatives (for example according to WO 01/049806), amine derivatives having fused aromatic systems (for example according to U.S. Pat. No. 5,061,569), the amine derivatives disclosed in WO 95/09147, monobenzoindenofluorenamines (for example according to WO 08/006449), dibenzoindenofluorenamines (for example according to WO 07/140847), spirobifluorenamines (for example according to WO 2012/034627, WO2014/056565), fluorenamines (for example according to EP 2875092, EP 2875699 and EP 2875004), spirodibenzopyranamines (e.g. EP 2780325) and dihydroacridine derivatives (for example according to WO 2012/150001).

(233) The device is correspondingly (according to the application) structured, contact-connected and finally hermetically sealed, since the lifetime of such devices is severely shortened in the presence of water and/or air.

(234) Additionally preferred is an organic electroluminescent device, characterized in that one or more layers are coated by a sublimation process. In this case, the materials are applied by vapour deposition in vacuum sublimation systems at an initial pressure of typically less than 10^{-5} mbar, preferably less than 10^{-6} mbar. It is also possible that the initial pressure is even lower or even higher, for example less than 10^{-7} mbar.

(235) Preference is likewise given to an organic electroluminescent device, characterized in that one or more layers are coated by the OVPD (organic vapour phase deposition) method or with the aid of a carrier gas sublimation. In this case, the materials are applied at a pressure between 10^{-5} mbar and 1 bar. A special case of this method is the OVJP (organic vapour jet printing) method, in which the materials are applied directly by a nozzle and thus structured (for example M. S.

Arnold et al., *Appl. Phys. Lett.* 2008, 92, 053301).

(236) Preference is additionally given to an organic electroluminescent device, characterized in that one or more layers are produced from solution, for example by spin-coating, or by any printing method, for example screen printing, flexographic printing, offset printing or nozzle printing, but more preferably LITI (light-induced thermal imaging, thermal transfer printing) or inkjet printing. For this purpose, soluble compounds are needed, which are obtained, for example, through suitable substitution.

(237) The organic electroluminescent device can also be produced as a hybrid system by applying one or more layers from solution and applying one or more other layers by vapour deposition. For example, it is possible to apply an emitting layer comprising a metal complex of the invention and a matrix material from solution, and to apply a hole blocker layer and/or an electron transport layer thereto by vapour deposition under reduced pressure.

(238) These methods are known in general terms to those skilled in the art and can be applied by those skilled in the art without difficulty to organic electroluminescent devices comprising compounds of formula (1) or the above-detailed preferred embodiments.

(239) The electronic devices of the invention, especially organic electroluminescent devices, are notable for one or more of the following surprising advantages over the prior art: 1. With the metal complexes of the invention, red, orange, yellow, green and blue emission is obtainable with high efficiency and lifetime. 2. The metal complexes of the invention have very good hydrolysis stability. In particular, hydrolysis stability is much better than in the case of complexes containing pyrazolylborate derivatives as ligands, but in which the ligands do not have polypodal bridging. The compounds of the invention are therefore also of very good suitability for processing from solution. The improved hydrolysis stability is also an advantage in the synthesis of the complexes. 3. Organic electroluminescent devices comprising the metal complexes of the invention as emitting materials have excellent efficiency and show oriented emission. The efficiency of emission is better than in comparable complexes that do not have a pyrazolylborate sub-ligand of the formula (2). This effect is particularly highly marked for metal complexes in which L^{sup.3} is a sub-ligand of the formula (2), i.e. those that have two sub-ligands of the formula (2). This effect is also particularly marked when the sub-ligand L^{sup.2} has a substituent of the formula (52) or (53), and especially when the sub-ligand L^{sup.3} simultaneously has a structure of the formula (2).

(240) These abovementioned advantages are not accompanied by a deterioration in the further electronic properties.

(241) The invention is illustrated in more detail by the examples which follow, without any intention of restricting it thereby. The person skilled in the art will be able to use the details given, without exercising inventive skill, to produce further electronic devices of the invention and hence to execute the invention over the entire scope claimed.

EXAMPLES

(242) The syntheses which follow, unless stated otherwise, are conducted under a protective gas atmosphere in dried solvents. The metal complexes are additionally handled with exclusion of light or under yellow light. The solvents and reagents can be purchased, for example, from Sigma-ALDRICH or ABCR. The respective figures in square brackets or the numbers quoted for individual compounds relate to the CAS numbers of the compounds known from the literature. In the case of compounds that can have multiple tautomeric forms, one tautomeric form is shown in a representative manner.

A: Synthesis of the Synthons S

Example S1

(243) ##STR00304##

(244) To a well-stirred mixture of 11.2 g (100 mmol) of 1H-pyrazole-4-boronic acid [763120-58-7], 28.3 g (100 mmol) of 2-bromoiodobenzene [583-55-1] and 21.2 g (200 mmol) of sodium carbonate in 300 ml of dioxane and 50 ml of water are added 1.63 g (2 mmol) of dichloro[1,1'-

bis(diphenylphosphino)ferrocene]palladium(II) x CH₂Cl₂.sub.2, and the mixture is then heated under reflux for 16 h. After cooling, 300 ml of saturated sodium chloride solution are added, and the organic phase is removed and dried over magnesium sulfate. The desiccant is filtered out of the mixture using a silica gel bed in a dioxane slurry, the filtrate is concentrated under reduced pressure and the residue is subjected to flash chromatography (Torrent automatic column system from A. Semrau). Yield: 12.0 g (54 mmol), 54%. Purity: about 95% by ¹H NMR.

(245) In an analogous manner, it is possible to prepare the following compounds:

(246) TABLE-US-00001 Ex. Reactants Product Yield S2 05

57% S3 07 51% S4 09

0 45% S5 50% S6

48% S7 52% S8

53% S9 0 45%

Example S30

(247) ##STR00321##

(248) A well-stirred mixture of 66.3 g (100 mmol) of 2,2'-[5''-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-[1,1':2',1'':3'',1''':2''',1''''-quinquephenyl]-4,4''''-diyl]bispyridine [1989597-72-9], 22.3 g (100 mmol) of S1, 21.2 g (200 mmol) of sodium carbonate, 1.23 g (3 mmol) of SPhos, 449 mg (2 mmol) of palladium(II) acetate, 300 ml of toluene, 100 ml of ethanol and 300 ml of water is heated under reflux for 16 h. After cooling, the mixture is extended with 300 ml of toluene, and the organic phase is removed, washed once with 500 ml of water and once with 500 ml of saturated sodium chloride solution and dried over magnesium sulfate. After the desiccant has been filtered off using a Celite bed in a toluene slurry and the solvent has been removed under reduced pressure, the residue is recrystallized from acetonitrile/ethyl acetate. Yield: 49.6 g (73 mmol), 73%. Purity: about 97% by ¹H NMR.

(249) In an analogous manner, it is possible to prepare the following compounds:

(250) TABLE-US-00002 Ex. Reactants Product Yield S31 70% S32 75% S33 72% S34 70% S35 0 67% S36 76% S37 78% S38 64% S39 77% S40 0 73% S41 69% S42 76%

Example S50

(251) ##STR00346##

(252) To a well-stirred mixture of 18.2 g (50 mmol) of 2,2'-(5-chloro-1,3-phenylene)bis[4,4,5,5-tetramethyl-1,3,2-dioxaborolane [1417036-49-7], 22.3 g (100 mmol) of S1 and 21.2 g (200 mmol) of sodium carbonate in 300 ml of toluene, 200 ml of ethanol and 300 ml of water are added 1.6 g (6 mmol) of triphenylphosphine and then 449 mg (2 mmol) of palladium(II) acetate, and the mixture is then heated under reflux for 24 h. After cooling, the toluene phase is removed and washed once with 300 ml of water and once with 300 ml of saturated sodium chloride solution, and then dried over magnesium sulfate. The mixture is filtered through a Celite bed in a toluene slurry, the toluene is removed under reduced pressure and the residue is recrystallized from acetonitrile/methanol. Yield: 12.7 g (32 mmol), 64%. Purity: about 95% by ¹H NMR.

Example S51

(253) ##STR00347##

(254) To a mixture of 19.8 g (50 mmol) of S50, 14.0 g (55 mmol) of bis(pinacolato)diborane [78183-34-3], 8.2 g (100 mmol) of sodium acetate and 50 g of glass beads (diameter 3 mm) in 300 ml of THE are successively added 534 mg (1.3 mmol) of SPhos and then 224 mg (1 mmol) of palladium(II) acetate, and the mixture is stirred under reflux for 16 h. After cooling, the reaction mixture is filtered through a Celite bed in a THE slurry. The filtrate is concentrated to dryness

under reduced pressure, and the residue is extracted by stirring in 200 ml of warm methanol. The product is filtered off with suction, washed twice with 50 ml each time of methanol and dried under reduced pressure. Yield: 23.0 g (47 mmol), 94%. Purity: about 95% by ¹H NMR.

Example S60

(255) ##STR00348##

(256) A well-stirred mixture of 48.8 g (100 mmol) of S51, 31.0 g (100 mmol) of 2-(2'-bromo[1,1'-biphenyl]-4-yl)pyridine [1374202-35-3], 21.2 g (200 mmol) of sodium carbonate, 1.23 g (3 mmol) of SPhos, 449 mg (2 mmol) of palladium(II) acetate, 300 ml of toluene, 100 ml of ethanol and 300 ml of water is heated under reflux for 16 h. After cooling, the mixture is extended with 300 ml of toluene, and the organic phase is removed, washed once with 500 ml of water and once with 500 ml of saturated sodium chloride solution and dried over magnesium sulfate. After the desiccant has been filtered off using a Celite bed in a toluene slurry and the solvent has been removed under reduced pressure, the residue is recrystallized from acetonitrile/ethyl acetate. Yield: 37.3 g (63 mmol), 63%. Purity: about 97% by ¹H NMR.

(257) In an analogous manner, it is possible to prepare the following compounds:

(258) TABLE-US-00003 Ex. Reactants Product Yield S61  0  66% S62   60% S63   61% S64   60% S65   64% S66  0  68% S67   68% S68   70% S69   66%

B: Synthesis of the Complexes

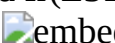
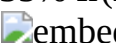
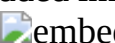
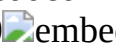
Example Ir(L30)

(259) ##STR00367##

(260) To a well-stirred solution of 6.79 g (10 mmol) S30 in 200 ml of dry benzonitrile are added, at 60° C., 8.27 g (30 mmol) of silver carbonate [534-16-7] and 50 g of glass beads (diameter 3 mm), and the mixture is stirred for 5 min. Then 3.22 g (10 mmol) of tris(acetonitrile)trichloroiridium(III) [85835-70-7] are added, and the reaction mixture is heated to 130° C. on a water separator for 6 h, in the course of which the acetonitrile is distilled off. The mixture is allowed to cool down to 60° C., 2.8 g (12 mmol) of 1,5-cyclooctanediyl(N,N-dimethylamine)(1H-pyrazolato-N^{sup.1})borane [125050-95-5] are added, and the mixture is stirred at 60° C. for 16 h. The benzonitrile is removed under reduced pressure, the residue is dissolved in 300 ml of dichloromethane (DCM), and then the DCM is distilled off on a rotary evaporator, with simultaneous dropwise addition of MeOH until the crude product crystallizes. The crude product is filtered off with suction, washed three times with 30 ml each time of MeOH and dried under reduced pressure. The crude product thus obtained is dissolved in 200 ml of dichloromethane and filtered through about 1 kg of silica gel in the form of a dichloromethane slurry (column diameter about 18 cm) with exclusion of air in the dark, leaving dark-coloured components at the start. The core fraction is cut out and concentrated on a rotary evaporator, with simultaneous continuous dropwise addition of MeOH until crystallization.

(261) After removal with suction, washing with a little MeOH and drying under reduced pressure, the orange product is purified further by continuous hot extraction five times with dichloromethane/acetonitrile 1:1 (v/v) (amount initially charged in each case about 200 ml, extraction thimble: standard Soxhlet thimbles made from cellulose from Whatman) with careful exclusion of air and light. The loss into the mother liquor can be adjusted via the ratio of dichloromethane (low boilers and good dissolvers):acetonitrile (high boilers and poor dissolvers). It should typically be 3-6% by weight of the amount used. Hot extraction can also be accomplished using other solvents or solvent mixtures such as toluene, xylene, ethyl acetate, butyl acetate, i-PrOH etc. Finally, the product is fractionally sublimed at 380-440° C. under high vacuum. Yield: 3.82 g (3.6 mmol), 36%. Purity: >99.9% by HPLC.

(262) In an analogous manner, it is possible to prepare the following compounds:

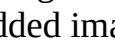
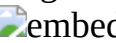
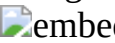
(263) TABLE-US-00004 Ex. Reactants Product Yield Ir(L31)  embedded image
 embedded image 24% Ir(L32) S32 [56953-94-7] 0  embedded image 35% Ir(L33)
 embedded image  embedded image 32% Ir(L34)  embedded image  embedded image 74%
 Ir(L35) S35 [125050-95-5]  embedded image 34% Ir(L36) S36 [125050-95-5]  embedded image
 33% Ir(L37) S37 [125050-95-5]  embedded image 29% Ir(L38) S38 [125050-95-5]
 embedded image 34% Ir(L39) S39 [125050-95-5]  embedded image 37% Ir(L40) S40 [125050-95-5] 0
 embedded image 43% Ir(L41) S41 [125050-95-5]  embedded image 45% Ir(L42) S42
 [125050-95-5]  embedded image 40%

Example Ir(L60)

(264) ##STR00383##

(265) To a well-stirred solution of 5.92 g (10 mmol) S60 in 200 ml of dry benzonitrile are added, at 60° C., 8.27 g (30 mmol) of silver carbonate [534-16-7] and 50 g of glass beads (diameter 3 mm), and the mixture is stirred for 5 min. Then 3.22 g (10 mmol) of tris(acetonitrile)trichloroiridium(III) [85835-70-7] are added, and the reaction mixture is heated to 130° C. on a water separator for 6 h, in the course of which the acetonitrile is distilled off. The mixture is allowed to cool down to 60° C., 4.98 g (25 mmol) of sodium [dimethylbis(1H-pyrazolato-N.sup.1)]borate [56953-94-7] are added, and the mixture is stirred at 60° C. for 16 h. The benzonitrile is removed under reduced pressure, the residue is dissolved in 300 ml of dichloromethane (DCM), and then the DCM is distilled off on a rotary evaporator, with simultaneous dropwise addition of MeOH until the crude product crystallizes. The crude product is filtered off with suction, washed three times with 30 ml each time of MeOH and then dried under reduced pressure. The crude product thus obtained is purified and sublimed as described in Example Ir(L30). Yield: 3.01 g (3 mmol), 30%. Purity: >99.9% by HPLC.

(266) In an analogous manner, it is possible to prepare the following compounds:

(267) TABLE-US-00005 Ex. Reactants Product Yield Ir(L61) S61 [16453-63-7]
 embedded image 29% Ir(L62) S62 [125050-95-5]  embedded image 34% Ir(L63) S63 [125050-95-5]
 embedded image 30% Ir(L64) S64 [125050-95-5]  embedded image 29% Ir(L65) S65
 [125050-95-5]  embedded image 37% Ir(L66) S66 [125050-95-5]  embedded image 35%
 Ir(L67) S67 [125050-95-5] 0  embedded image 27% Ir(L68) S68 [56953-94-7]  embedded image
 31% Ir(L69) S69 [56953-94-7]  embedded image 33%

Hydrolysis Stability of the Complexes:

(268) For processing from solution (see example: Production of the OLEDs, solution-processed devices), the complexes must have very good hydrolysis stability since the residual water present in the solvent can otherwise result in hydrolytic breakdown. Even hydrolytic breakdown to a slight degree can have a very adverse effect on the component properties of the OLEDs with regard to efficiency and in particular lifetime. To verify hydrolysis stability, 15 mg of the complex are dissolved in 0.75 ml of DMSO-d₆, 50 µl of H.sub.2O are added, and the mixture is stored at 60° C. for 8 h. Thereafter, a ¹H NMR spectrum (1024 scans) is recorded and compared with the ¹H NMR spectrum of the complex in dry DMSO-d₆, the solution likewise having been stored at 60° C. for 8 h. Hydrolysis is perceptible by the occurrence of new signals. These can be assigned to the free ligand, hydrolysis products of the ligand and aquo complexes. The results are compiled in Table 1 below.

(269) TABLE-US-00006 TABLE 1 Hydrolysis properties Observation of Ex. Complex hydrolysis
 H1 Ref-Ir1 yes see (Table 4) H2 Ir(L31) no H3 Ir(L60) no

Example: Production of the OLEDs

1) Vacuum-Processed Devices

(270) OLEDs of the invention and OLEDs according to the prior art are produced by a general method according to WO 2004/058911, which is adapted to the circumstances described here (variation in layer thickness, materials used). In the examples which follow, the results for various OLEDs are presented. Cleaned glass plaques (cleaning in Miele laboratory glass washer, Merck

Extran detergent) coated with structured ITO (indium tin oxide) of thickness 50 nm are pretreated with UV ozone for 25 minutes (PR-100 UV ozone generator from UVP) and, within 30 min, for improved processing, coated with 20 nm of PEDOT:PSS (poly(3,4-ethylenedioxythiophene) poly(styrenesulfonate), purchased as CLEVIOS™ P VP AI 4083 from Heraeus Precious Metals GmbH Deutschland, spun on from aqueous solution) and then baked at 180° C. for 10 min. These coated glass plaques form the substrates to which the OLEDs are applied. The OLEDs basically have the following layer structure: substrate/hole injection layer 1 (HIL1) consisting of HTM doped with 5% NDP-9 (commercially available from Novaled), 20 nm/hole transport layer 1 (HTL1) consisting of HTM, 160 nm for blue devices, 220 nm for green/yellow devices/electron blocker layer (EBL)/emission layer (EML)/hole blocker layer (HBL)/electron transport layer (ETL)/optional electron injection layer (EIL) and finally a cathode. The cathode is formed by an aluminium layer of thickness 100 nm.

(271) For vacuum-processed OLEDs, all materials are applied by thermal vapour deposition in a vacuum chamber. In this case, the emission layer always consists of at least one matrix material (host material) and an emitting dopant (emitter) which is added to the matrix material(s) in a particular proportion by volume by co-evaporation. Details given in such a form as M1:M2:Ir(L2) (55%:35%:10%) mean here that the material M1 is present in the layer in a proportion by volume of 55%, M2 in a proportion by volume of 35% and Ir(L1) in a proportion by volume of 10%. Analogously, the electron transport layer may also consist of a mixture of two materials. The exact structure of the OLEDs can be found in Table 2. The materials used for production of the OLEDs are shown in Table 5.

(272) The OLEDs are characterized in a standard manner. For this purpose, the electroluminescence spectra, the current efficiency (measured in cd/A), the power efficiency (measured in lm/W) and the external quantum efficiency (EQE, measured in percent) as a function of luminance, calculated from current-voltage-luminance characteristics (IUL characteristics) assuming Lambertian emission characteristics, and also the lifetime are determined. The electroluminescence spectra are determined at a luminance of 1000 cd/m.^{sup.2}, and the CIE 1931 x and y colour coordinates are calculated therefrom. The lifetime LT80 is defined as the time after which the luminance drops to 80% of the starting luminance in the course of operation with a constant current of 40 mA/cm.^{sup.2}.

(273) Use of Compounds of the Invention as Emitter Materials in Phosphorescent OLEDs

(274) One use of the compounds of the invention is as phosphorescent emitter materials in the emission layer in OLEDs. The results for the OLEDs are collated in Table 3.

(275) TABLE-US-00007 TABLE 2 Structure of the OLEDs EBL EML HBL ETL Ex. thickness thickness thickness thickness D1 EBM1 M1:Ir(L30) ETM1 ETM1:ETM2 10 nm (83%:17%) 10 nm (50%:50%) 30 nm 30 nm D2 EBM1 M1:M2:Ir(L30) ETM1 ETM1:ETM2 10 nm (50%:38%:12%) 10 nm (50%:50%) 30 nm 30 nm D3 EBM1 M1:Ir(L31) ETM1 ETM1:ETM2 5 nm (78%:22%) 10 nm (50%:50%) 30 nm 30 nm D4 EBM2 M1:M2:Ir(L67) ETM1 ETM1:ETM2 10 nm (50%:38%:12%) 10 nm (50%:50%) 30 nm 30 nm D5 EBM2 M1:M2:Ir(L68) ETM1 ETM1:ETM2 10 nm (50%:38%:12%) 10 nm (50%:50%) 30 nm 30 nm

(276) TABLE-US-00008 TABLE 3 Results for the vacuum-processed OLEDs EQE (%) Voltage (V) CIE x/y LT80 (h) Ex. 1000 cd/m.^{sup.2} 1000 cd/m.^{sup.2} 1000 cd/m.^{sup.2} 40 mA/cm.^{sup.2} D1 23.6 3.1 0.36/0.61 220 D2 23.9 2.9 0.36/0.62 250 D3 21.4 3.3 0.18/0.35 90 D4 23.0 3.0 0.38/0.60 360 D5 26.3 3.1 0.39/0.60 330

Solution-Processed Devices:

From Soluble Functional Materials of Low Molecular Weight

(277) The iridium complexes of the invention may also be processed from solution and lead therein to OLEDs which are much simpler in terms of process technology compared to the vacuum-processed OLEDs, but nevertheless have good properties. The production of such components is based on the production of polymeric light-emitting diodes (PLEDs), which has already been

described many times in the literature (for example in WO 2004/037887). The structure is composed of substrate/ITO/hole injection layer (60 nm)/interlayer (20 nm)/emission layer (60 nm)/hole blocker layer (10 nm)/electron transport layer (40 nm)/cathode. For this purpose, substrates from Technoprint (soda-lime glass) are used, to which the ITO structure (indium tin oxide, a transparent conductive anode) is applied. The substrates are cleaned in a cleanroom with DI water and a detergent (Deconex 15 PF) and then activated by a UV/ozone plasma treatment. Thereafter, likewise in a cleanroom, a 20 nm hole injection layer is applied by spin-coating. The required spin rate depends on the degree of dilution and the specific spin-coater geometry. In order to remove residual water from the layer, the substrates are baked on a hotplate at 200° C. for 30 minutes. The interlayer used serves for hole transport. In the present case, an HL-X from Merck is used. The interlayer may alternatively also be replaced by one or more layers which merely have to fulfil the condition of not being leached off again by the subsequent processing step of EML deposition from solution. For production of the emission layer, the triplet emitters of the invention are dissolved together with the matrix materials in toluene or chlorobenzene. The typical solids content of such solutions is between 16 and 25 g/l when, as here, the layer thickness of 60 nm which is typical of a device is to be achieved by means of spin-coating. The solution-processed devices of type 1 contain an emission layer composed of M3:M4:IrL (20%:60%:20%), and those of type 2 contain an emission layer composed of M3:M4:IrLa:IrLb (30%:34%:30%:6%); in other words, they contain two different Ir complexes. The emission layer is spun on in an inert gas atmosphere, argon in the present case, and baked at 160° C. for 10 min. Vapour-deposited above the latter are the hole blocker layer (10 nm ETM1) and the electron transport layer (40 nm ETM1 (50%)/ETM2 (50%)) (vapour deposition systems from Lesker or the like, typical vapour deposition pressure 5×10⁻⁶ mbar). Finally, a cathode of aluminium (100 nm) (high-purity metal from Aldrich) is applied by vapour deposition. In order to protect the device from air and air humidity, the device is finally encapsulated and then characterized. The OLED examples cited have not yet been optimized. Table 4 summarizes the data obtained.

(278) TABLE-US-00009 TABLE 4 Results with materials processed from solution Voltage EQE (%) (V) LT50 (h) Emitter 1000 1000 1000 Ex. Device cd/m² cd/m² CIE x/y cd/m²
Sol-D1 Ir(L33) 21.3 4.4 0.30/0.63 270000 Type 1 Sol-D2 Ir(L63) 19.8 4.3 0.65/0.34 300000
Ir(L213) Type 2

(279) TABLE-US-00010 TABLE 5 Structural formulae of the materials used     
000102

Claims

1. A compound of the formula (1) ##STR00403## where the symbols used are as follows: L^{sup.1} is a sub-ligand of the formula (2) which coordinates to the iridium via the 1 two nitrogen atoms identified by * and which is bonded to V via the dotted bond, ##STR00404## where: A is the same or different at each instance and is CR or N, where not more than one A group per ring is N; R^{sup.B} is the same or different at each instance and is F, OR^{sup.1}, a straight-chain alkyl group having 1 to 10 carbon atoms or a branched or cyclic alkyl group having 3 to 10 carbon atoms, where the alkyl group in each case may be substituted by one or more R^{sup.1} radicals, or an aromatic or heteroaromatic ring system which has 5 to 24 aromatic ring atoms and may be substituted in each case by one or more R^{sup.1} radicals; at the same time, the two R^{sup.B} radicals together may also form a ring system; provided that R^{sup.B} is not an unsubstituted pyrazole; L^{sup.2} is a bidentate, monoanionic sub-ligand which coordinates to the iridium via one carbon atom and one nitrogen atom or via two carbon atoms; L^{sup.3} is a bidentate, monoanionic sub-ligand which coordinates to the iridium via one carbon atom and one nitrogen atom or via two carbon atoms, or is a sub-ligand of the formula (2) which may be the same as or different from L¹;

V is a group of the formula (3), where the dotted bonds each represent the bonds to the sub-ligands L.sup.1, L.sup.2 and L.sup.3, ##STR00405## X.sup.1 is the same or different at each instance and is CR or N; X.sup.2 is the same or different at each instance and is CR or N, or two adjacent X.sup.2 groups together are NR, O or S, thus forming a five-membered ring; or two adjacent X.sup.2 groups together are CR or N when one of the X.sup.3 groups in the cycle is N, thus forming a five-membered ring; with the proviso that not more than two adjacent X.sup.2 groups in each ring are N; X.sup.3 is C at each instance in one cycle or one X.sup.3 group is N and the other X.sup.3 group in the same cycle is C, where the X.sup.3 groups in the three cycles may be selected independently, with the proviso that two adjacent X.sup.2 groups together are CR or N when one of the X.sup.3 groups in the cycle is N; R is the same or different at each instance and is H, D, F, Cl, Br, I, N(R.sup.1).sub.2, OR.sup.1, SR.sup.1, CN, NO.sub.2, COOH, C(=O)N(R.sup.1).sub.2, Si(R.sup.1).sub.3, B(OR.sup.1).sub.2, C(=O)R.sup.1, P(=O)(R.sup.1).sub.2, S(=O)R.sup.1, S(=O).sub.2R.sup.1, OSO.sub.2R.sup.1, a straight-chain alkyl group having 1 to 20 carbon atoms or an alkenyl or alkynyl group having 2 to 20 carbon atoms or a branched or cyclic alkyl group having 3 to 20 carbon atoms, where the alkyl, alkenyl or alkynyl group may in each case be substituted by one or more R.sup.1 radicals and where one or more nonadjacent CH.sub.2 groups may be replaced by Si(R.sup.1).sub.2, C=O, NR.sup.1, O, S or CONR.sup.1, or an aromatic or heteroaromatic ring system which has 5 to 40 aromatic ring atoms and may be substituted in each case by one or more R.sup.1 radicals; at the same time, two R radicals together may also form a ring system; R.sup.1 is the same or different at each instance and is H, D, F, Cl, Br, I, N(R.sup.2).sub.2, OR.sup.2, SR.sup.2, CN, NO.sub.2, Si(R.sup.2).sub.3, B(OR.sup.2).sub.2, C(=O)R.sup.2, P(=O)(R.sup.2).sub.2, S(=O)R.sup.2, S(=O).sub.2R.sup.2, OSO.sub.2R.sup.2, a straight-chain alkyl group having 1 to 20 carbon atoms or an alkenyl or alkynyl group having 2 to 20 carbon atoms or a branched or cyclic alkyl group having 3 to 20 carbon atoms, where the alkyl, alkenyl or alkynyl group may in each case be substituted by one or more R.sup.2 radicals and where one or more nonadjacent CH.sub.2 groups may be replaced by Si(R.sup.2).sub.2, C=O, NR.sub.2, O, S or CONR.sub.2, or an aromatic or heteroaromatic ring system which has 5 to 40 aromatic ring atoms and may be substituted in each case by one or more R.sup.2 radicals; at the same time, two or more R.sup.1 radicals together may form a ring system; R.sup.2 is the same or different at each instance and is H, D, F or an aliphatic, aromatic and/or heteroaromatic organic radical having 1 to 20 carbon atoms, in which one or more hydrogen atoms may also be replaced by F; at the same time, the three bidentate ligands L.sup.1, L.sup.2 and L.sup.3, apart from by the V group, may also be closed by a further bridge to form a cryptate.

2. The compound of claim 1, wherein the group of the formula (3) is selected from the structures of the formulae (4b) to (7b) ##STR00406##

3. The compound of claim 1, wherein the group of the formula (3) is selected from the structures of the formulae (4c) and (5c): ##STR00407##

4. The compound of claim 1, wherein the sub-ligand L.sup.1 has a structure of the formula (2a) ##STR00408## where the substituents R adjacent to the coordinating nitrogen atom are the same or different at each instance and are selected from the group consisting of H, D, F, methyl, ethyl and phenyl.

5. The compound of claim 1, wherein the sub-ligand L.sup.1 has a structure of the formula (2c) ##STR00409##

6. The compound of claim 1, wherein the substituents R.sup.B are the same or different at each instance and are selected from the group consisting of OR.sup.1 where R.sup.1 is an alkyl group having 1 to 5 carbon atoms, a straight-chain alkyl group having 1 to 5 carbon atoms, a branched or cyclic alkyl group having 3 to 6 carbon atoms, an aryl group which has 6 to 10 carbon atoms and may be substituted by one or more R.sup.1 radicals, or a heteroaryl group which has 5 to 10 aromatic ring atoms and may be substituted by one or more R.sup.1 radicals, where the two R.sup.B radicals together may also form a ring system; or in that the two R.sup.B radicals together

- with the boron atom to which they are bonded represent a group of one of the following formulae (B-1) to (B-8): ##STR00410## wherein the dotted lines represent the bonds to ##STR00411## of Formula (2); provided that R^{sup}.B is not an unsubstituted pyrazole.
7. The compound of claim 1, wherein L^{sup}.3 represents a sub-ligand of the formula (2), where the sub-ligands L^{sup}.1 and L^{sup}.3 may be the same or different.
8. The compound of claim 1, wherein L^{sup}.2 has one carbon atom and one nitrogen atom as coordinating atoms.
9. The compound of claim 1, wherein the sub-ligand L^{sup}.2 is a structure of one of the formulae (L-1) or (L-2), and in that the sub-ligand L^{sup}.3, when it coordinates to the iridium via one carbon atom and one nitrogen atom or two carbon atoms, is a structure of one of the formulae (L-1) and (L-2) ##STR00412## where the dotted bond represents the bond of the sub-ligand to the group of the formula (3) and the other symbols used are as follows: CyC is the same or different at each instance and is a substituted or unsubstituted aryl or heteroaryl group which has 5 to 14 aromatic ring atoms and coordinates in each case to the metal via a carbon atom and which is bonded to CyD via a covalent bond; CyD is the same or different at each instance and is a substituted or unsubstituted heteroaryl group which has 5 to 14 aromatic ring atoms and coordinates to the metal via a nitrogen atom or via a carbene carbon atom and which is bonded to CyC via a covalent bond; at the same time, two or more substituents together may form a ring system.
10. The compound of claim 9, wherein CyC represents an aryl or heteroaryl group which has 6 to 13 aromatic ring atoms and coordinates to the metal via a carbon atom and which may be substituted by one or more R radicals and which is bonded to CyD via a covalent bond, and in that CyD represents a heteroaryl group which has 5 to 13 aromatic ring atoms and coordinates to the metal via an uncharged nitrogen atom or via a carbene carbon atom and which may be substituted by one or more R radicals and which is bonded to CyC via a covalent bond.
11. The compound of claim 1, wherein the sub-ligand L^{sup}.2 is substituted by a substituent of the formula (52) or (53) ##STR00413## where the dotted bond indicates the linkage of the group to the sub-ligand L^{sup}.2 and, in addition: R' is the same or different at each instance and is selected from the group consisting of H, D, F, CN, a straight-chain alkyl group having 1 to 10 carbon atoms in which one or more hydrogen atoms may also be replaced by D or F, or a branched or cyclic alkyl group having 3 to 10 carbon atoms in which one or more hydrogen atoms may also be replaced by D or F, or an alkenyl group having 2 to 10 carbon atoms in which one or more hydrogen atoms may also be replaced by D or F; at the same time, two adjacent R' radicals or two R' radicals on adjacent phenyl groups together may also form a ring system; or two R' on adjacent phenyl groups together are a group selected from O and S, such that the two phenyl rings together with the bridging group are a dibenzofuran or dibenzothiophene, and the further R' are as defined above; n is 0, 1, 2, 3, 4 or 5.
12. A process for preparing a compound of claim 1, wherein an iridium salt is reacted with a ligand precursor in the presence of a pyrazolylborate synthon and in the presence of a halogen scavenger.
13. A formulation comprising at least one compound of claim 1 and at least one solvent and/or at least one further organic or inorganic compound.
14. An electronic device, oxygen sensitizer, photoinitiator, or photocatalyst comprising at least one compound of claim 1.
15. An electronic device comprising at least one compound of claim 1.
16. The electronic device of claim 15, wherein it is an organic electroluminescent device and the compound of claim 1 is present together with a matrix material in one or more emitting layers.
17. The compound of claim 1, wherein R^{sup}.B is not pyrazole.
-